SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Robotics, Cognition, Intelligence

# Design and Control of an Aerodynamic Surface-Enhanced Multirotor

William Constantin Rosenhahn

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I confirm that this master’s thesis is my own work and I have documented all sources and material used.

Munich, 01.12.2025 William Constantin Rosenhahn

# Acknowledgments

"I would like to thank Pablo Kirby for developing the improved platform variant and supporting the experimental validation, and for the collaborative work environment."

# Abstract

This thesis presents the design, modeling, and experimental validation of an X-wing hybrid aerial platform that combines quadrotor vertical take-off and landing (VTOL) capability with fixed-wing aerodynamic efficiency. The platform features four wings arranged in an X-configuration, providing passive lift that reduces required rotor thrust during forward flight.

A comprehensive dynamics model incorporating aerodynamic forces and moments is developed and integrated with an Incremental Nonlinear Dynamic Inversion (INDI) control architecture. The controller requires no explicit aerodynamic model and treats wing-generated forces as bounded disturbances, enabling robust trajectory tracking across the entire flight envelope from hover to high-speed cruise.

Experimental validation was conducted in a motion capture arena with a 2.5 kg prototype. Two configurations were tested: NACA 0015 airfoils at 45° dihedral and AG25 airfoils at 20° dihedral. Thrust characterization yielded the mapping $T = 5.15 \times 10^{-6} \cdot \mathrm{RPM}^2 - 0.090 \times 10^{-2} \cdot \mathrm{RPM}$ (N) with $R^2 = 0.997$. Flight tests demonstrated stable, accurate tracking up to $8\,\mathrm{m\,s^{-1}}$ with RMS position errors below 0.24 m, sustaining centripetal accelerations up to 2.8g, demonstrating stable control under strong unmodeled aerodynamic loads and indicative rotor unloading.

Electrical power was directly measured in outdoor flight using synchronized current and voltage telemetry. Results show a measurable reduction in mechanical power demand with increasing airspeed—up to 33% at $10\mathrm{m\,s^{-1}}$—confirming that aerodynamic lift from the fixed surfaces translates to tangible energy savings while maintaining full control authority.

# Contents

# 1. Introduction

Conventional multirotors achieve unmatched agility and control simplicity but remain power-inefficient in sustained forward flight because all lift must be generated by rotor thrust. Fixed-wing aircraft, in contrast, exploit aerodynamic lift for high efficiency but cannot hover. Hybrid vertical-take-off-and-landing (VTOL) systems attempt to merge both regimes, yet they require additional actuators, complex transition control, and elaborate modeling.

This thesis investigates a minimal hybrid concept: a quadrotor augmented with fixed aerodynamic surfaces. The objective is to examine whether such a configuration can achieve measurable lift-induced power relief while retaining the full controllability of a multirotor and without increasing mechanical complexity.

While several studies have demonstrated that fixed aerodynamic surfaces can reduce power consumption in forward flight, none have examined how such unmodeled aerodynamic forces affect trajectory-tracking accuracy or control robustness when using a disturbance-rejection controller such as Incremental Nonlinear Dynamic Inversion (INDI). This gap is central: aerodynamic lift may improve efficiency but also introduces nonlinear couplings that could degrade control performance. The present work investigates this interaction directly.

Improving forward-flight efficiency is meaningful only if control performance is preserved. Aerodynamic surfaces alter force and moment coupling, which may burden or destabilize a conventional multirotor controller. Thus, efficiency and control robustness must be evaluated together.

INDI is chosen because its incremental formulation rejects slowly varying disturbances without requiring an aerodynamic model. Aerodynamic lift from fixed wings constitutes precisely such a disturbance. This thesis therefore tests the hypothesis that INDI maintains tracking accuracy under significant, unmodeled aerodynamic lift, eliminating the need for model-based compensation.

## 1.1. Research question

The central research question is:

*Can an unmodified INDI controller maintain accurate trajectory tracking on a* *quadrotor with significant aerodynamic lift, and does the resulting rotor unloading translate into measurable electrical power savings?*

To answer this, the work combines modeling, design, and experimental validation. A six-degree-of-freedom dynamics model with simplified aerodynamic terms is derived. A fully 3D-printed "X-wing" quadrotor platform is built to match this model. The controller architecture employs a cascaded INDI structure with a linear outer loop for position control and a geometric-based inner INDI loop for attitude control, implemented in C++ using ROS 1 and validated through motion-capture experiments, as well as outdoor flights with synchronized power logging.

The scope of this work is intentionally limited: no aerodynamic model is used in the controller, and no transition or hybrid flight regime is considered. Instead, the study isolates the interaction between fixed-wing lift and an unmodified INDI controller in forward flight, without introducing gain scheduling or aerodynamic compensation.

## 1.2. Contributions

The contributions of this thesis are:

- **Platform design**: A structurally symmetric X-wing quadrotor integrating four fixed wings into the rotor arms for passive lift generation.
- **Dynamics model**: A minimal 6-DoF model including quadratic lift and drag suitable for control and simulation.
- **Control architecture**: A cascaded INDI structure with a linear outer loop for position control and a geometric-based inner INDI loop for attitude control that compensates aerodynamic disturbances without explicit modeling.
- **Experimental validation**: System identification, model validation, and indoor motion-capture trials as well as outdoor flights with synchronized power logging.
- **Efficiency assessment**: Measured rotor-power reduction up to 33% at $10\,\mathrm{m\,s^{-1}}$ compared with hover, confirming that passive lift directly reduces electrical power consumption.

These measurements provide the first quantitative demonstration of real-flight power savings on an aerodynamic surface-enhanced quadrotor under INDI control.

This investigation provides a reproducible reference for future research on aerodynamic surface-enhanced multirotors and hybrid UAV control without reliance on aerodynamic parameter identification.

## 1.3. Thesis organization

The following chapter surveys prior work in quadrotor control, hybrid VTOL architectures, and aerodynamic augmentation, highlighting the absence of systematic evaluation of control robustness under passive aerodynamic lift. Chapter 2 reviews related work and motivates the chosen design. Chapter 3 derives the model. Chapter 4 details the platform. Chapter 5 presents the controller. Chapters 6 and 7 describe the setup and experiments. Chapter 8 concludes.

# 2. Related Work

This chapter reviews existing literature in three areas relevant to multirotors equipped with fixed aerodynamic surfaces: (i) quadrotor control approaches and their disturbance-rejection properties, (ii) control strategies for hybrid VTOL and fixed-wing/transitioning platforms, and (iii) prior work on aerodynamic augmentation of multirotors. Together, these strands expose the gap addressed in this thesis: the absence of a systematic evaluation of a disturbance-rejection controller such as INDI on a multirotor that generates substantial aerodynamic lift.

## 2.1. Quadrotor Control Approaches

Modern multirotor control is dominated by geometric controllers and incremental nonlinear dynamic inversion (INDI).

**Geometric control on SE(3).** Geometric controllers operate directly on the configuration space of the quadrotor, avoiding attitude singularities and ensuring well-defined error metrics on SO(3) [**Lee2010**; GLL15]. These approaches provide precise tracking of aggressive trajectories and are now standard in many high-performance flight stacks. Their performance, however, depends on reasonably accurate modeling of rotational dynamics and control effectiveness.

**Incremental Nonlinear Dynamic Inversion (INDI).** INDI provides robustness to model errors by controlling incremental changes in angular acceleration rather than relying on a full dynamic model [SCM10; SCC16]. The method uses measured angular rates, finite-difference angular accelerations, and motor-speed feedback to estimate control effectiveness, exploiting the time-scale separation between fast actuators and slower vehicle dynamics. Quasi-steady aerodynamic forces—such as rotor drag, prop-wake asymmetries, and external disturbances—appear only as slowly varying terms and are rejected through the incremental structure. INDI has demonstrated strong disturbance rejection in aggressive flight [Tzo+21] and under significant aerodynamic loading without explicit aerodynamic identification.

**Summary.** The quadrotor literature establishes that INDI can tolerate substantial unmodeled aerodynamic forces, provided actuator dynamics are fast and control effectiveness is known. This makes INDI a promising candidate for platforms with additional passive aerodynamic surfaces—but this has not been systematically evaluated.

## 2.2. Hybrid VTOL and Fixed-Wing/Transition Control

Hybrid VTOL platforms combine multirotor hovering with fixed-wing cruise capability. They fall into two major categories:

**Tilt-rotor and tilt-wing aircraft.** Tilt-rotor concepts reorient propulsion units mechanically [**Hartmann2017**; **Bauersfeld2021**]. Control design requires accurate aerodynamic models, explicit transition logic, and coordinated actuator scheduling. Mode switches between hover and forward flight introduce nonlinear couplings that must be explicitly handled.

**Tailsitters and transition-capable platforms.** Tailsitters retain fixed actuators but operate over radically different aerodynamic regimes [Oos+17; TK22]. Control architectures typically blend multirotor and fixed-wing control laws, with transition management depending on angle-of-attack estimation and aerodynamic model interpolation [Da24]. Gains must adapt to the current flight regime to ensure stability during transition.

**Summary.** All hybrid VTOL concepts share two traits: (1) they require explicit aerodynamic modeling, and (2) they require mode-dependent control architectures (gain scheduling, blending, or transition logic). This complexity stands in contrast to the simplicity of a pure multirotor controller, raising the question whether a simpler design—adding fixed wings without changing the control structure—can achieve useful efficiency benefits.

## 2.3. Aerodynamically Augmented Multirotors

Between pure multirotors and full VTOL systems lies a class of platforms that integrate fixed passive aerodynamic surfaces while retaining standard quadrotor actuation.

**Small-wing augmentation studies.** Dawkins and DeVries [DD18] demonstrated $\approx$ 35 % power reduction at 3–5 $\mathrm{m\,s^{-1}}$ using small wings on a quadrotor, with the native PID controller unchanged. Xiao et al. [XQa20] achieved $\approx 50\,\%$ electrical power reduction at $15\,\mathrm{m\,s^{-1}}$ on a $1.2\,\mathrm{kg}$ platform, also without modifying the baseline controller.

These studies show that fixed wings can significantly reduce propulsive power. However, both works evaluate only power consumption, not control performance. They provide no analysis of:

- trajectory tracking accuracy,
- disturbance rejection,
- coupling effects between aerodynamics and attitude dynamics,
- controller robustness limits.

**Partial-transition tailsitter studies.** Beyond small-wing augmentation, tailsitter platforms provide the closest analogue to the concept investigated in this thesis. In tailsitter flight prior to transition, the vehicle experiences significant aerodynamic lift and drag while still relying on multirotor-style attitude control laws. Several studies [Zho+17; Oos+17; TK22; Da24] show that conventional or slightly adapted multirotor controllers can stabilise such vehicles in regimes where aerodynamic forces dominate thrust-generated forces, without requiring an explicit aerodynamic model. These works demonstrate that robust attitude controllers—especially those leveraging incremental or disturbance-rejection principles—can tolerate substantial, unmodeled aerodynamic coupling during partial-transition flight. Although tailsitters differ mechanically from the fixed-wing quadrotor considered here, they provide empirical evidence that multirotor control architectures remain viable under strong aerodynamic loading. This insight directly motivates the present investigation: if multirotor-style control can handle the aerodynamic complexity of tailsitter pre-transition flight, a disturbance-rejecting controller such as INDI may likewise maintain tracking performance on a quadrotor equipped with fixed wings.

**Summary.** The literature demonstrates that (i) fixed wings provide measurable efficiency gains, and (ii) multirotor controllers can survive some aerodynamic influence. But no prior work quantifies how aerodynamic lift and drag affect tracking accuracy when a disturbance-rejection controller such as INDI is used without modification.

## 2.4. Gap and Scope

Across quadrotor control, VTOL architecture, and aerodynamic augmentation studies, the missing link is clear:

*No existing work systematically examines the control performance—especially tracking accuracy and disturbance rejection—of an unmodified INDI controller on a multirotor that generates significant aerodynamic lift.*

Prior works prove energy savings, but none analyze how unmodeled quasi-steady aerodynamic forces interact with the control structure.

This thesis addresses that gap by experimentally evaluating an INDI-controlled quadrotor equipped with fixed wings across a range of forward-flight speeds, quantifying both aerodynamic effects and control performance.

# 3. Dynamics Model

This chapter formulates the six-degree-of-freedom dynamics used throughout the thesis. The model serves two purposes: (i) to quantify the aerodynamic forces and moments that act as disturbances on the multirotor, and (ii) to establish the reference against which the INDI controller must reject these disturbances. As discussed in Chapter 2, INDI-based controllers have demonstrated strong robustness to aerodynamic loading in both classical quadrotors and tailsitter pre-transition regimes, enabling the present work to use a simplified aerodynamic description without compromising control fidelity. The aerodynamic terms introduced here are used only for offline analysis; the controller in Chapter 5 deliberately ignores them.

## 3.1. List of Symbols

Table 3.1.: Symbols used in the dynamics model.

| Symbol | Description | Unit |
|---|---|---|
| $\mathcal{W}$ | World (inertial) frame (ENU) | — |
| $\mathcal{B}$ | Body-fixed frame at CoG | — |
| $\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$ | Body-frame unit vectors | — |
| $\mathbf{p}$ | Position in world frame | m |
| $\mathbf{v}$ | Velocity in world frame | m/s |
| $R$ | Rotation matrix $\mathcal{B} \to \mathcal{W}$ | — |
| $\boldsymbol{\omega}$ | Angular velocity in body frame | rad/s |
| $\mathbf{g}$ | Gravity vector $[0, 0, -g]^\top$ | $\mathrm{m/s^2}$ |
| $m$ | Vehicle mass | kg |
| $J$ | Inertia matrix in body frame | $\mathrm{kg \cdot m^2}$ |
| $\mathbf{F}_g$ | Gravitational force | N |
| $f_i$ | Thrust produced by rotor $i$ | N |
| $\mathbf{u}$ | Control input vector $[f_1, f_2, f_3, f_4]^\top$ | N |
| $\omega_i$ | Angular speed of rotor $i$ | rad/s |

Continued on next page

Table 3.1 – continued from previous page

| **Symbol** | **Description** | **Unit** |
|---|---|---|
| $k_t$ | Thrust coefficient | $\mathrm{N{\cdot}s^2{\cdot}rad^{-2}}$ |
| $b_i$ | Static bias in thrust model for rotor $i$ | N |
| $k_Q$ | Torque coefficient | $\mathrm{N{\cdot}m{\cdot}s^2{\cdot}rad^{-2}}$ |
| $Q_i$ | Reaction torque of rotor $i$ | N·m |
| $c_i$ | Prop rotation sign ($\pm 1$) | — |
| $\mathbf{v}_w$ | Wind velocity in world frame | m/s |
| $\mathbf{v}_a$ | Airspeed in body frame | m/s |
| $V$ | Airspeed magnitude | m/s |
| $\alpha$ | Angle of attack | rad |
| $\beta$ | Sideslip angle | rad |
| $\rho$ | Air density | $\mathrm{kg/m^3}$ |
| $q$ | Dynamic pressure $= \frac{1}{2}\rho V^2$ | $\mathrm{N/m^2}$ |
| $S$ | Wing reference area | $\mathrm{m^2}$ |
| $b$ | Wing span | m |
| $AR$ | Aspect ratio $b^2/S$ | — |
| $a_\alpha$ | Lift-curve slope | $\mathrm{rad^{-1}}$ |
| $a_\beta$ | Side-force slope (or $C_{Y_\beta}$) | $\mathrm{rad^{-1}}$ |
| $C_L, C_D, C_Y$ | Lift, drag, side-force coefficients | — |
| $C_{D0}$ | Zero-lift drag coefficient | — |
| $k$ | Induced drag factor | — |
| $\mathbf{F}$ | Total force in world frame | N |
| $\mathbf{F}_T$ | Rotor thrust force in world frame | N |
| $\mathbf{F}_{aero}$ | Aerodynamic force in body frame | N |
| $\boldsymbol{\tau}$ | Total moment in body frame | N·m |
| $\boldsymbol{\tau}_T$ | Rotor-generated moment | N·m |
| $\boldsymbol{\tau}_{aero}$ | Aerodynamic moment | N·m |
| $\mathbf{r}_i$ | Arm vector to rotor $i$ | m |
| $\mathbf{r}_w$ | Aerodynamic center offset | m |
| $\hat{\mathbf{t}}_i$ | Rotor $i$ thrust direction unit vector | — |
| $G$ | Allocation matrix (force–moment mapping) | — |
| $\hat{\mathbf{v}}$ | Airspeed unit vector | — |
| $\hat{\mathbf{z}}_L$ | Lift direction unit vector | — |
| $\hat{\mathbf{y}}_S$ | Side-force direction unit vector | — |

## 3.2. Frames, States, Inputs

We use a world (inertial) frame $\mathcal{W}$ with ENU convention and a body-fixed frame $\mathcal{B}$ at the CoG. Gravity is $\mathbf{g} = [0,\, 0,\, -g]^\top$ in $\mathcal{W}$.

State:

$$(\mathbf{p}, \mathbf{v}, R, \boldsymbol{\omega}) \in \mathbb{R}^3 \times \mathbb{R}^3 \times SO(3) \times \mathbb{R}^3,$$

where $\mathbf{p}, \mathbf{v}$ are position/velocity in $\mathcal{W}$, $R$ maps $\mathcal{B} \to \mathcal{W}$, and $\boldsymbol{\omega}$ is the body angular rate in $\mathcal{B}$.

Control input (per rotor $i \in \{1, \ldots, 4\}$):

$$\mathbf{u} = [f_1,\, f_2,\, f_3,\, f_4]^\top, \qquad f_i \geq 0,$$

with nominal thrust along $+\mathbf{e}_3$ of $\mathcal{B}$. Motor dynamics and allocation are covered in Chapter 5.

**Wind and airspeed.** Let $\mathbf{v}_w \in \mathbb{R}^3$ be the wind in $\mathcal{W}$. Body-frame airspeed:

$$\mathbf{v}_a \;=\; R^\top(\mathbf{v} - \mathbf{v}_w) \;\in\; \mathbb{R}^3, \quad V \;=\; \|\mathbf{v}_a\|.$$

Angle of attack and sideslip:

$$\alpha = \arctan 2(-v_{a,z}, v_{a,x}), \qquad \beta = \arcsin\left(\frac{v_{a,y}}{V}\right).$$

## 3.3. Forces in the World Frame

Total force:

$$\mathbf{F} \;=\; m\mathbf{g} \;+\; \mathbf{F}_T \;+\; R\,\mathbf{F}_{aero}. \tag{3.1}$$

### 3.3.1. Gravity

$\mathbf{F}_g = m\mathbf{g}$.

### 3.3.2. Rotor Thrust

Each rotor $i$ produces a thrust $f_i$ along its (possibly tilted) direction $\hat{\mathbf{t}}_i$ in $\mathcal{B}$. The net thrust in $\mathcal{W}$ is

$$\mathbf{F}_T = R \sum_{i=1}^{4} f_i\, \hat{\mathbf{t}}_i. \tag{3.2}$$

For small fixed tilts (e.g., $\pm 5°$) we also provide the common approximation $\mathbf{F}_T \approx (\sum_i f_i)\, R\mathbf{e}_3$; all tilt effects are embedded in the allocation matrix $G$ used for moments.

### 3.3.3. Aerodynamic Forces on Fixed Surfaces

We assume the wing span aligns with $\mathbf{e}_y$, so lift lies in the $(x, z)$ plane. The definitions of $\hat{\mathbf{z}}_L$ and $\hat{\mathbf{y}}_S$ are undefined when $\hat{\mathbf{v}}$ is parallel to the respective cross-product axis (e.g., $\hat{\mathbf{v}} \parallel \mathbf{e}_y$ or $\hat{\mathbf{v}} \parallel \mathbf{e}_z$); in implementation we fall back to the last valid direction or a small-angle approximation.

Dynamic pressure $q = \frac{1}{2}\rho V^2$, total reference area $S$.

Unit aerodynamic directions in $\mathcal{B}$ (from $\mathbf{v}_a$):

$$\hat{\mathbf{v}} = \frac{\mathbf{v}_a}{V}, \quad \hat{\mathbf{z}}_L = \frac{\hat{\mathbf{v}} \times \mathbf{e}_y}{\|\hat{\mathbf{v}} \times \mathbf{e}_y\|}, \quad \hat{\mathbf{y}}_S = \frac{\mathbf{e}_z \times \hat{\mathbf{v}}}{\|\mathbf{e}_z \times \hat{\mathbf{v}}\|},$$

so that lift is perpendicular to the flow in the body $x$–$z$ plane, drag opposes $\hat{\mathbf{v}}$, and side force is lateral.

Aerodynamic force in $\mathcal{B}$:

$$\mathbf{F}_{aero} = \underbrace{qS\, C_L(\alpha)\, \hat{\mathbf{z}}_L}_{\text{lift}} - \underbrace{qS\, C_D(\alpha)\, \hat{\mathbf{v}}}_{\text{drag}} + \underbrace{qS\, C_Y(\beta)\, \hat{\mathbf{y}}_S}_{\text{side}}. \tag{3.3}$$

The aerodynamic forces in (3.3) are used only for off-line analysis; they are not used in the controller.

**Controller relevance.** These aerodynamic forces are *not* modeled in the controller; they are treated as disturbances. INDI cancels their quasi-steady contribution via incremental updates (Chapter 5).

## 3.4. Moments in the Body Frame

Total moment:

$$\boldsymbol{\tau} = \boldsymbol{\tau}_T + \boldsymbol{\tau}_{aero}. \tag{3.4}$$

### 3.4.1. Rotor-Generated Moments

Let $\mathbf{r}_i$ be the arm vector from CoG to rotor $i$, and $c_i \in \{+1, -1\}$ the prop rotation sign (reaction torque direction). Here $c_i \in \{+1, -1\}$ encodes the rotor spin direction (positive for CCW viewed from above). With small motor tilt handled by per-rotor thrust directions $\hat{\mathbf{t}}_i$,

$$\boldsymbol{\tau}_{\text{arms}} = \sum_{i=1}^{4} \mathbf{r}_i \times (f_i\, \hat{\mathbf{t}}_i), \tag{3.5}$$

$$\boldsymbol{\tau}_{\text{react}} = \sum_{i=1}^{4} c_i\, Q_i\, \mathbf{e}_3, \qquad Q_i \approx k_Q\, \omega_i^2, \tag{3.6}$$

so that $\boldsymbol{\tau}_T = \boldsymbol{\tau}_{\text{arms}} + \boldsymbol{\tau}_{\text{react}}$. In nominal modeling $\hat{\mathbf{t}}_i = \mathbf{e}_3$; with $\pm 5^\circ$ motor tilt we precompute $\hat{\mathbf{t}}_i$ and absorb it in the allocation matrix $G$ (see Chapter 5).

### 3.4.2. Aerodynamic Moments

Wing forces act at an effective aerodynamic center offset $\mathbf{r}_w$ (in $\mathcal{B}$). We lump unsteady effects into a disturbance:

$$\boldsymbol{\tau}_{aero} \;=\; \mathbf{r}_w \times \mathbf{F}_{aero} \;+\; \boldsymbol{\tau}_{aero,dist}. \tag{3.7}$$

As with forces, the controller does not model $\boldsymbol{\tau}_{aero}$ explicitly; INDI compensates the quasi-steady part incrementally.

## 3.5. Equations of Motion

Rigid-body dynamics:

$$\begin{aligned} \dot{\mathbf{p}} &= \mathbf{v}, &(3.8)\\ \dot{\mathbf{v}} &= \tfrac{1}{m}\big(m\mathbf{g} + \mathbf{F}_T + R\,\mathbf{F}_{aero}\big), &(3.9)\\ \dot{R} &= R[\boldsymbol{\omega}]_\times, &(3.10)\\ J\dot{\boldsymbol{\omega}} &= -\boldsymbol{\omega} \times J\boldsymbol{\omega} + \boldsymbol{\tau}, &(3.11) \end{aligned}$$

with inertia $J$ in $\mathcal{B}$ and $[\cdot]_\times$ the skew operator.

## 3.6. Thrust and Allocation (Static Mapping)

For simulation and identification we use static rotor maps

$$f_i \;=\; k_t\, \omega_i^2 + b_i, \qquad Q_i \;=\; k_Q\, \omega_i^2,$$

and a (geometry-dependent) allocation matrix $G \in \mathbb{R}^{4\times 4}$ such that

$$\begin{bmatrix} f_T \\ \boldsymbol{\tau} \end{bmatrix} \;=\; G\,\mathbf{f}, \qquad \mathbf{f} = [f_1, f_2, f_3, f_4]^\top, \quad f_T = \mathbf{1}^\top \mathbf{f}.$$

Motor tilt is embedded in $G$ via $\hat{\mathbf{t}}_i$ and the arm geometry. Chapter 5 details the inverse mapping and constraints.

### 3.6.1. Rotor Power Scaling

Understanding how rotor power depends on rotational speed is essential for evaluating the efficiency of hybrid multirotor–fixed-wing configurations. In hover, this relationship can be derived from momentum theory, which models the rotor as an ideal actuator disk accelerating air downward to produce thrust.

For a rotor in hover, the thrust $T$ is generated by imparting a downward momentum to the air mass flow $\dot{m}$:

$$T = \dot{m}v_i = (\rho A v_i)v_i = \rho A v_i^2,$$

where $\rho$ is the air density, $A$ is the rotor disk area, and $v_i$ is the induced velocity through the disk. The corresponding mechanical power required to sustain this thrust is

$$P = T v_i = \rho A v_i^3.$$

Eliminating $v_i$ using the first equation yields the classical induced power expression from actuator-disk theory:

$$P = T\sqrt{\frac{T}{2\rho A}} = \frac{T^{3/2}}{\sqrt{2\rho A}}$$

(see [Fed23]).

If blade geometry and collective pitch remain constant, blade-element theory shows that the thrust coefficient $C_T$ remains approximately constant, implying

$$T \propto \Omega^2,$$

where $\Omega$ is the rotor angular velocity. Substituting this into the induced power relation gives the cubic power law

$$P \propto \Omega^3.$$

Thus, reducing rotor speed by 20% results in a power reduction of roughly $(0.8)^3 \approx 0.51$, or about 49% lower power consumption—an important efficiency benefit when part of the lift is provided by wings.

This relationship strictly applies to hover or near-hover conditions, assuming uniform inflow and negligible profile and parasitic losses. In forward or edgewise flight, blade-element momentum theory (BEMT) or CFD analyses are required for accurate power prediction, but the cubic scaling remains a useful first-order engineering approximation for small deviations around hover.

The cubic law provides a theoretical baseline later compared against measured electrical-power data from outdoor experiments (Chapter 7), enabling direct verification of aerodynamic efficiency gains.

## 3.7. Assumptions and Simplifications

- Quasi-steady aerodynamics.
- No explicit prop-wash/wing interference; its net effect is absorbed in the disturbance terms and identified coefficients.
- Rigid airframe, symmetric mass properties about $\mathcal{B}$ axes.
- Fixed-pitch rotors; flapping and gyroscopic coupling are neglected in the baseline model.

## 3.8. Summary and Outlook

The presented six-degree-of-freedom equations define a consistent reference for the vehicle's translational and rotational dynamics. Aerodynamic terms were included only to quantify expected disturbance magnitudes—order of lift, drag, and pitching moments—so that the later control analysis can assess whether these effects are sufficiently slow and quasi-steady to be rejected by INDI. No aerodynamic coefficients or force models are used online; they serve purely for interpretation and comparison.

With the governing relationships and notation established, the next step is to translate this abstract model into a physical platform whose geometry and mass distribution match the assumed parameters. The following chapter therefore details the mechanical design and manufacturing of the X-wing quadrotor, showing how the chosen dimensions, airfoil, and dihedral angles realize the modeled reference frame in practice.

# 4. Platform Design

The platform was designed to isolate the aerodynamic effects of fixed lifting surfaces on a quadrotor without introducing new actuators or changing the control structure. The result is an X-wing configuration in which each rotor arm forms a fixed lifting surface. The design goal was to achieve measurable lift at moderate forward speeds (5–15 $\mathrm{m\,s^{-1}}$) while preserving the agility and controllability of a conventional multirotor.

## 4.1. Configuration and Layout

The vehicle retains the standard X-shaped quadrotor geometry but extends each arm into a symmetric wing section, producing a cross-shaped planform. This geometry satisfies three constraints:

- **Dynamic symmetry**: Equal inertias and aerodynamic properties in roll and pitch enable identical control gains across both axes.
- **Passive lift generation**: Each wing contributes lift proportional to its forward velocity component without interfering with rotor control authority.
- **Mechanical simplicity**: No additional servos or morphing structures are introduced; all aerodynamic effects are passive.

The $\pm 45°$ alternating dihedral and anhedral angles maintain directional symmetry, ensuring that any heading can serve as "forward" flight. Although such extreme dihedral is aerodynamically suboptimal compared with planar wings, it simplifies the control problem by avoiding preferred directions and limiting coupling between roll and pitch.

The wings employ a NACA 0015 symmetric profile with 0.25 m chord and 0.45 m span, yielding an aspect ratio of $\approx 1.8$. These dimensions were selected to ensure that at $\approx$ 10–12 $\mathrm{m\,s^{-1}}$ the predicted lift equals or slightly exceeds the vehicle weight, based on first-order thin-airfoil theory. The design therefore enables significant lift without requiring impractically large structures.

## 4.2. Structural Design and Manufacturing

Each wing consists of a 3D-printed aerodynamic shell reinforced by two 10 mm carbon spars. The hybrid construction provides sufficient bending stiffness while keeping individual wing mass below 180 g. Structural resonance tests indicated the first bending mode well above the control bandwidth, confirming negligible coupling to attitude dynamics.

The central frame clamps the eight spars from the four wings and houses all electronics. Motors are mounted on integrated endcaps that also serve as landing feet, providing 5° outward motor tilt for yaw authority. The total diagonal motor-to-motor distance is 1.1 m, yielding a compact yet aerodynamically relevant planform.

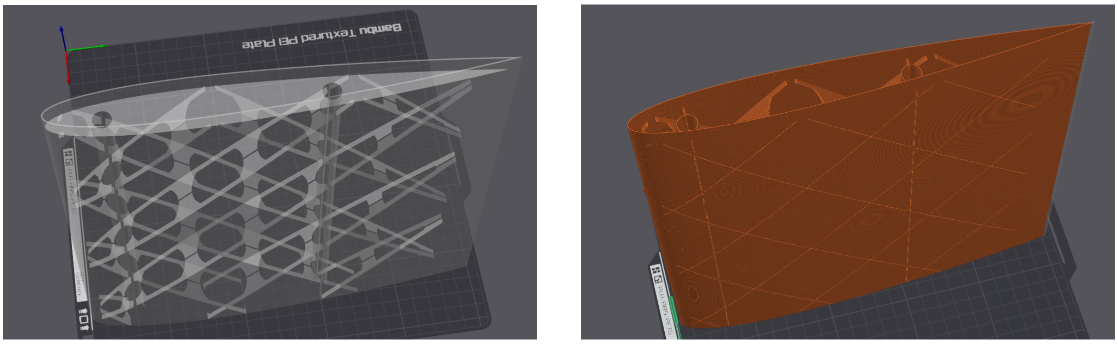

Figure 4.1.: Wing structure showing the hollow internal geometry with channels for carbon fiber spars (left) and printed aerodynamic shell (right). Each wing weighs approximately 176 g including carbon reinforcement.

## 4.3. Aerodynamic Estimates

At $12\,\mathrm{m\,s^{-1}}$, using $\rho = 1.225\,\mathrm{kg\,m^{-3}}$, $S = 0.318\,\mathrm{m^2}$, $C_L = 0.97$, $C_D = 0.025$, the theoretical lift and drag are:

$$L \approx 27.2\,\mathrm{N}, \quad D \approx 0.7\,\mathrm{N}.$$

This corresponds to $\approx 110\,\%$ of the vehicle's 2.5 kg weight, suggesting that in steady forward flight the rotors primarily counteract drag and provide control authority. Although these values are first-order estimates and neglect prop-wash interaction, they provide a reference for disturbance magnitudes that the INDI controller must compensate.

More accurate modeling would require CFD or wind-tunnel validation; however, for the scope of this study the analytical approximation suffices to show that aerodynamic

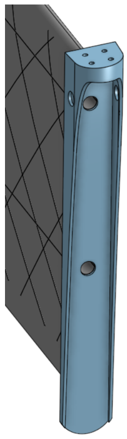

Figure 4.2.: Wing endcap component serving as motor mount and landing foot. This multi-functional part positions the motor ahead of the leading edge with $\pm 5^\circ$ tilt for yaw control and extends downward to protect the wing trailing edge during landing.

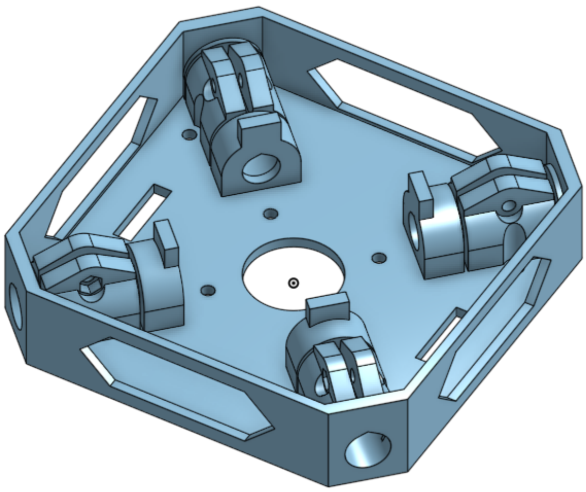

Figure 4.3.: Central core structure that clamps the carbon fiber spars. Two identical cores are used: one houses the ESC, flight controller, and batteries (top), while the other serves as a structural clamp for the lower spars (bottom).

lift is on the same order as vehicle weight and therefore cannot be ignored in flight analysis, even if it remains unmodeled in control.

A detailed comparison of the NACA 0015 baseline airfoil against an improved AG25 airfoil, including aerodynamic performance metrics and their correlation with flight test results, is presented in Chapter 7.

## 4.4. Propulsion and Actuation

The propulsion system uses T-MOTOR VELOX V2808 motors with HQProp 7×3.5×3 propellers, producing up to 22 N static thrust per motor. The total thrust-to-weight ratio exceeds 3, allowing aggressive maneuvers and ensuring that aerodynamic loads never saturate actuator authority.

Motors are tilted outward by 5° to enhance yaw torque. The corresponding geometry is embedded in the allocation matrix used by the controller; no additional coupling compensation is required. Static thrust characterization (Chapter 7, Section 7.1) confirms a quadratic thrust–RPM relation with $R^2 = 0.997$, validating consistency between analytical assumptions and measured performance.

The complete airframe mass, including avionics and batteries, is $\approx 2.5$ kg. The center of gravity lies at the geometric center, aligned with the intersection of the four wing roots, minimizing trim moments during hover.

## 4.5. Summary

The X-wing configuration provides a mechanically simple and dynamically symmetric platform in which aerodynamic forces are large enough to challenge the controller. The design therefore fulfills the thesis objective: to expose an INDI-controlled quadrotor to significant unmodeled aerodynamic effects without introducing additional actuation or transition mechanisms.

A second airframe variant was later developed to improve aerodynamic efficiency in forward flight. This refined configuration reduced the dihedral from ±45° to ±20° and replaced the NACA 0015 airfoil with an AG25 profile. The lower dihedral reduces projected area loss and induced drag, while the AG25's cambered geometry increases lift-to-drag ratio at typical cruise Reynolds numbers. The aerodynamic coefficient and agility analyses in Chapter 7 include both configurations, while the outdoor power-efficiency validation employs the refined AG25 design to quantify the performance of the final, optimized prototype.

# 5. Control Architecture

The controller aims to maintain precise trajectory tracking on a multirotor whose aerodynamic forces are unmodeled but significant. The design therefore follows a minimal philosophy: *compensate measured effects rather than predict them*. The architecture consists of a cascaded INDI structure with a geometric-based outer loop for position control and an INDI inner loop for attitude control. Aerodynamic forces from the fixed wings are never estimated or modeled; they are treated as slowly varying disturbances measured implicitly through the IMU.

## 5.1. Overview

The overall control stack operates in three layers:

- **Reference generation** – produces dynamically feasible position, velocity, and yaw trajectories using differential-flatness polynomials.
- **Geometric-based outer INDI loop** – converts position/velocity errors into desired acceleration and attitude commands, with incremental compensation for aerodynamic forces.
- **INDI inner loop** – tracks the desired angular rates by incrementally adjusting motor thrusts based on measured angular accelerations.

This separation allows the aerodynamic effects—lift, drag, and induced moments—to appear only as disturbances in the measured accelerations, where they are cancelled incrementally rather than modeled.

## 5.2. Reference Generation

The quadrotor dynamics are differentially flat with respect to position and yaw. In forward-flight experiments, the desired yaw rate was computed from a coordinated-turn condition

$$\dot{\psi}_d \approx \frac{a_{y,d}}{V_d \cos\theta_d},$$

aligning the vehicle's $x$-axis with its velocity vector and minimizing sideslip.

## 5.3. Geometric Outer Loop

The outer loop computes a commanded acceleration

$$\mathbf{a}_c = \mathbf{K}_p(\mathbf{p}_d - \mathbf{p}) + \mathbf{K}_v(\dot{\mathbf{p}}_d - \dot{\mathbf{p}}) + \ddot{\mathbf{p}}_d - \mathbf{g}, \tag{5.1}$$

which is converted to a desired attitude $R_d$ and collective thrust $f_{\text{coll}}$ via

$$f_{\text{coll}} = m\,\mathbf{a}_c^\top (R\mathbf{e}_3). \tag{5.2}$$

To account for unmodeled aerodynamic lift, the measured specific-force residual between expected rotor thrust and IMU acceleration is incrementally added:

$$\mathbf{a}_{\text{corr}} = R\left(\frac{f_T}{m}\mathbf{e}_3 - \mathbf{a}_{\text{IMU}}\right). \tag{5.3}$$

The corrected acceleration command becomes $\mathbf{a}_c \leftarrow \mathbf{a}_c + \mathbf{a}_{\text{corr}}$. This outer INDI layer, based on geometric control principles, acts as a real-time aerodynamic feedforward: it cancels any steady lift or drag without identifying parameters.

## 5.4. INDI Inner Loop

The inner loop enforces the commanded angular acceleration using incremental updates

$$\Delta\mathbf{u} = B^{-1}(\dot{\omega}_d - \dot{\omega}_k), \tag{5.4}$$

where $B$ is the local control-effectiveness matrix obtained from static identification. Because aerodynamic moments vary slowly relative to the 500 Hz control rate, their difference $\Delta\mathbf{d} \approx 0$ over each step, and thus their influence cancels.

This property—rejection of quasi-steady disturbances—makes INDI well suited for aerodynamic surface-enhanced vehicles, where modeling the unsteady flow is impractical.

## 5.5. Implementation Details

- **Software stack**: The control architecture is based on the Agilicious framework [FKa22] from the University of Zurich, adapted for INDI-based control and implemented in ROS 1 (Noetic) on a Jetson Orin Nano, communicating via MAVLink with a Kakute H7 flight controller running modified Betaflight firmware.

- **Update rates**: Outer loop 300 Hz; inner loop 300 Hz; IMU 8 kHz; motor commands sent at 8 kHz DShot.
- **Sensor feedback**: Position and attitude from Vicon motion capture at 200 Hz; angular rates and accelerations from the flight-controller IMU.
- **Filtering**: Outer-loop filter at 300 Hz with 10 Hz cut-off frequency for INDI residuals.

## 5.6. Stability and Design Logic

No explicit aerodynamic model enters the control law; stability arises from the incremental formulation. Because both outer and inner loops respond only to *changes* in measured acceleration, any quasi-steady aerodynamic component (e.g., lift from wings) vanishes from the control error. This design directly tests the thesis hypothesis: that accurate control is achievable on an aerodynamically augmented multirotor without aerodynamic parameter identification.

## 5.7. Summary

The implemented controller combines the disturbance rejection of INDI at both control levels with the geometric control structure of a conventional multirotor. Its novelty lies not in new mathematics but in the demonstration that this cascaded INDI architecture remains stable and precise under strong, unmodeled aerodynamic loads. The following chapter details the experimental setup used to verify this claim.

# 6. Experimental Setup

The experimental setup was designed to validate the control concept on a physical platform with accurate ground-truth feedback and synchronized onboard telemetry. The objectives were (i) to verify that the INDI controller maintains stable flight under aerodynamic loading and (ii) to record sufficient data for quantitative assessment of tracking accuracy and thrust behavior.

## 6.1. Hardware Configuration

The test vehicle is the X-wing quadrotor described in Chapter 4 with a total mass of 2.5 kg. Each T-MOTOR VELOX V2808 motor equipped with HQProp 7×3.5×3 propellers produces up to 22 N static thrust, giving a thrust-to-weight ratio of 3.6. The motor layout and rotation directions follow the conventional “X” configuration (Figure 6.1).

The aerodynamic surfaces follow a NACA 0015 profile with 0.45 m span and 0.25 m chord at $\pm 45^\circ$ dihedral/anhedral. The total projected wing area is 0.318 $m^2$. The center of gravity coincides with the geometric center, minimizing trim moments.

Two identical 6 S 1400 mA h Li-Po batteries are connected in parallel, yielding 2.8 A h effective capacity. The avionics stack consists of a Kakute H7 v1.5 flight controller and a Jetson Orin Nano (8 GB) companion computer communicating via UART over MAVLink. The flight controller runs modified Betaflight firmware capable of reporting motor RPM and battery voltage through MAVLink for logging.

## 6.2. Platform Variants

Two geometrically similar but aerodynamically distinct airframe variants were tested. The baseline platform used $\pm 45^\circ$ dihedral with NACA 0015 symmetric airfoils and served for all initial control and agility experiments. The improved platform employed $\pm 20^\circ$ dihedral and AG25 cambered airfoils to enhance lift efficiency and reduce induced drag. Both share identical propulsion, avionics, and controller tuning, enabling direct aerodynamic comparison. The improved variant was used exclusively for the outdoor efficiency measurements, representing the final design iteration derived from the indoor coefficient analysis.

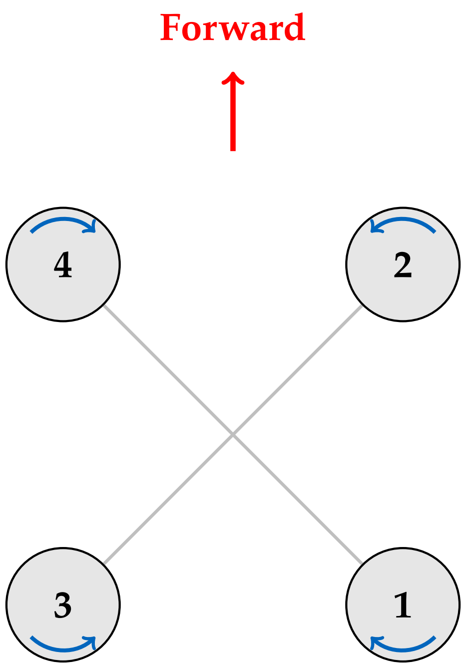


Figure 6.1.: Motor configuration and numbering convention. Blue arrows indicate rotation direction.

Figure 6.2.: The experimental X-wing quadrotor platform in the Vicon arena.

## 6.3. Thrust Characterization

Static thrust measurements were performed on a custom test stand using a six-axis Rokubi Mini force–torque sensor (Figure 6.3). To ensure realistic battery voltage sag matching actual flight conditions, all four motors were powered simultaneously from the flight battery (6 S 2 P configuration) with only one motor mounted on the force sensor. Motors were driven through the same Betaflight/DShot interface used in flight to ensure identical timing and response characteristics. This methodology guarantees that the measured RPM-to-thrust relationship accurately reflects the thrust produced during flight for a given motor command, accounting for realistic electrical loading and voltage drop. The measured mapping is used throughout the experiments for thrust estimation from motor telemetry. Full characterization methodology and results are presented in Section 7.1.

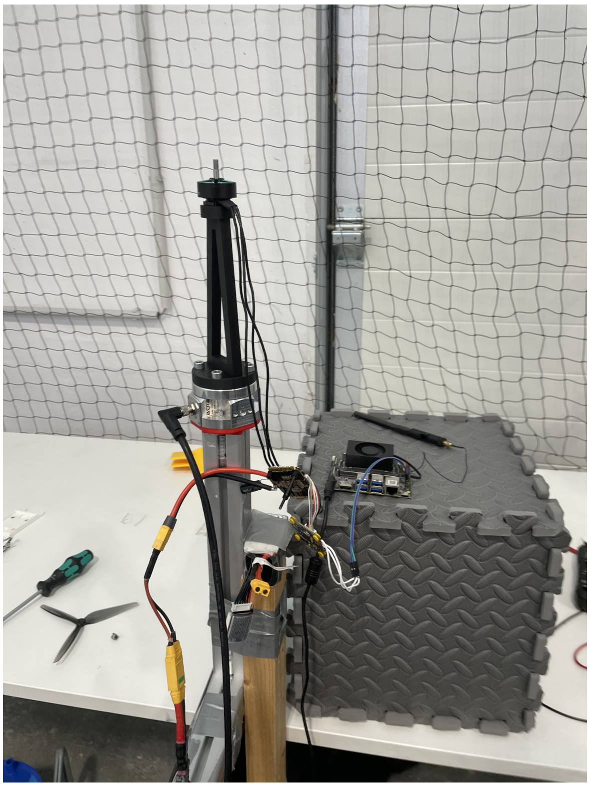

Figure 6.3.: Static thrust characterization test stand with motor mounted on Rokubi Mini force–torque sensor. All four motors were powered simultaneously from the flight battery to replicate realistic voltage sag, with one motor on the sensor.

## 6.4. Measurement and Logging

Ground-truth position and attitude were provided by Vicon motion-capture systems operating at 200 Hz. For agility trials, the $7 \times 8$ m Munich arena was used; for efficiency and high-speed tests, the $60 \times 45$ m AIDA Hall at Reutlingen University (Figure 6.4). Both datasets were timestamp-synchronized with onboard telemetry and merged in rosbag format.

Logged variables included:

- Position, velocity, attitude (Vicon)
- Angular rates and linear accelerations (IMU)
- Motor RPM and control commands
- Battery voltage

These channels provided all necessary signals for thrust estimation, trajectory-tracking evaluation, and disturbance-rejection analysis.

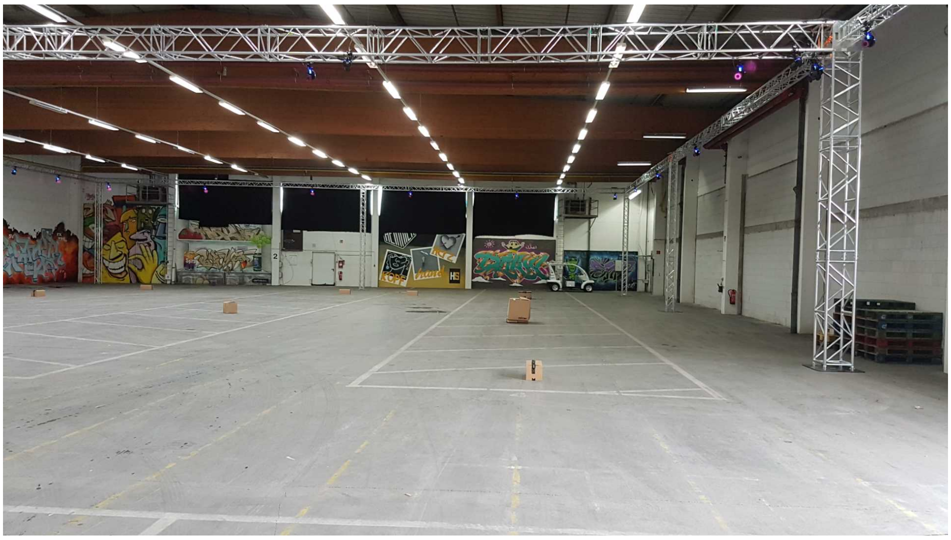

Figure 6.4.: AIDA Hall test facility used for efficiency trials (image credit: Reutlingen University [Reu24]).

## 6.5. Experimental Protocol

Each flight consisted of an autonomous take-off, hover stabilization, and a pre-programmed trajectory (step or circular). Manual override from the ground station was available for safety. Each dataset was verified for timing consistency and outliers before analysis.

## 6.6. Data Quality and Uncertainty

Sensor noise levels were assessed via static logs: IMU acceleration noise $\approx 0.004\,\mathrm{m\,s^{-2}}$ rms; Vicon position uncertainty $\pm 0.1\,\mathrm{mm}$; timestamp jitter $< 20\,\mathrm{ms}$. No measurable bias drift affected the evaluation interval ($\leq 1\,\mathrm{min}$ per flight). These figures ensure that the reported tracking errors in Chapter 7 exceed measurement noise by at least one order of magnitude.

## 6.7. Outdoor Power Measurement Setup

Electrical power was measured in outdoor flight using the flight controller's onboard voltage and current sensors. The Kakute H7 flight controller continuously sampled battery voltage and current draw at 1 kHz, logging the data via Betaflight's blackbox system alongside motor RPM, IMU data, and GPS telemetry.

Data were post-processed to compute instantaneous electrical power $P = VI$ at each sample, where $V$ is the battery voltage in volts and $I$ is the total current draw in amperes. To reduce high-frequency measurement noise while preserving transient behavior, the power signal was smoothed using a 1000-sample moving-average filter, corresponding to approximately 1 s averaging window at the 1 kHz logging rate.

This configuration enables direct measurement of total system electrical power consumption, including motors, flight controller, and all onboard electronics. Combined with synchronized motor RPM telemetry and GPS-based velocity measurements, these data allow quantitative comparison of power demand across different flight regimes (hover, cruise, acceleration) and direct validation of momentum theory predictions for rotor power scaling.

## 6.8. Summary

The experimental setup provides synchronized, high-fidelity data for validating the control approach. It achieves sufficient accuracy to resolve sub-decimeter trajectory errors and verify that the INDI controller can stabilize and track trajectories under unmodeled aerodynamic disturbances. This configuration allows direct comparison of electrical-power consumption between hover, level, and forward flight, enabling quantitative efficiency analysis.

# 7. Experiments and Evaluation

This chapter validates the control approach and aerodynamic performance of the X-wing platform through a series of experimental campaigns. The experiments address four key objectives:

1. Identify the static thrust-to-RPM mapping for control and energy analysis (Munich, thrust stand).
2. Evaluate trajectory-tracking performance and agility of the INDI controller (Munich, flight arena).
3. Quantify aerodynamic forces and validate computational predictions (AIDA indoor flight tests).
4. Measure direct electrical power consumption under realistic flight conditions (outdoor tests).

## 7.1. Static Thrust Identification

The thrust-characterization procedure described in Section 6.3 produced 975 measurements under the 6 S 2 P power configuration. A quadratic model was fitted to the zero-offset-corrected data:

$$T = 5.15 \times 10^{-8}\,\mathrm{RPM}^2 - 0.09 \times 10^{-3}\,\mathrm{RPM}, \tag{7.1}$$

with coefficient of determination $R^2 = 0.997$ (Figure 7.1).

The quadratic dependence confirms the classical blade-element scaling $T \propto \Omega^2$ and validates the assumption used in the control allocation. This mapping is used throughout all flight experiments to translate desired thrust into commanded motor speeds and to estimate thrust from measured RPM when reconstructing aerodynamic forces.

To interface with the low-level motor controller, a linear mapping between command and RPM was identified (Figure 7.2). This provides a simple feedforward relationship for allocating thrust commands to ESC duty cycles.

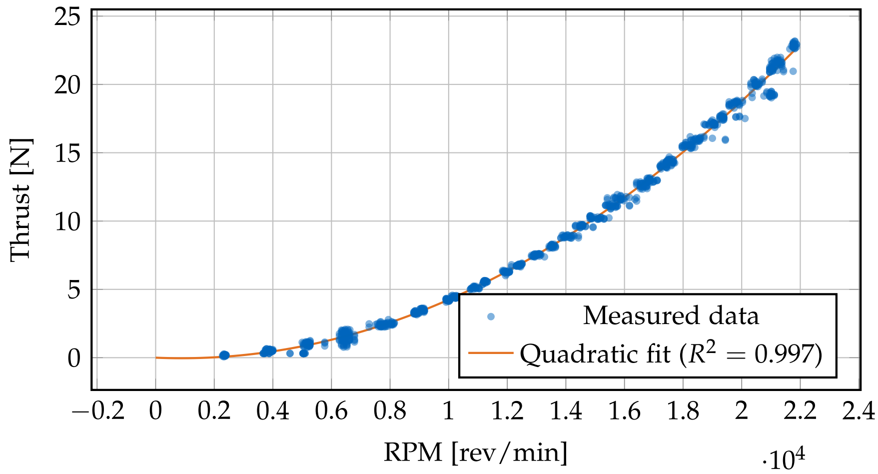


Figure 7.1.: Static thrust versus RPM with quadratic fit after zero-offset correction. The close agreement ($R^2 = 0.997$) confirms the $T \propto \Omega^2$ scaling used in control and analysis.

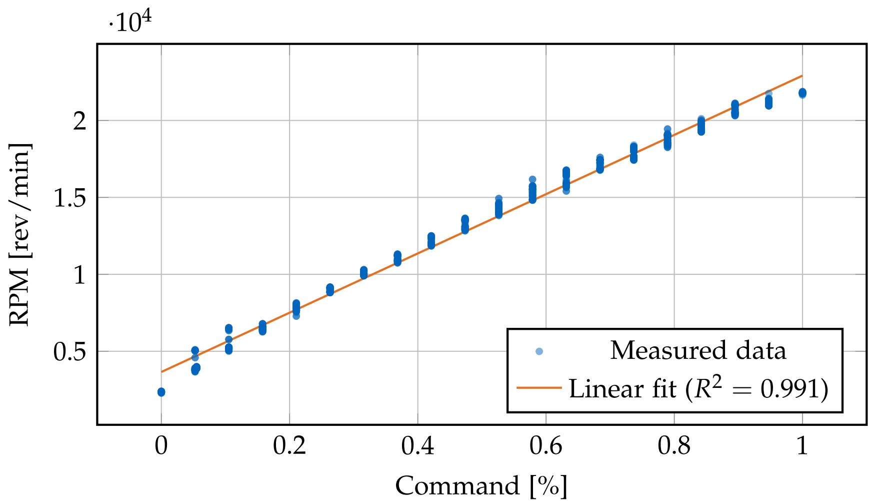


Figure 7.2.: Static mapping between motor command and RPM. The near-linear relationship simplifies thrust allocation and allows RPM prediction from control inputs.

For completeness, Figure 7.3 shows the direct mapping between motor command and thrust. This relationship is used only for sanity checks; all subsequent analysis relies on the RPM-based thrust model in (7.1).

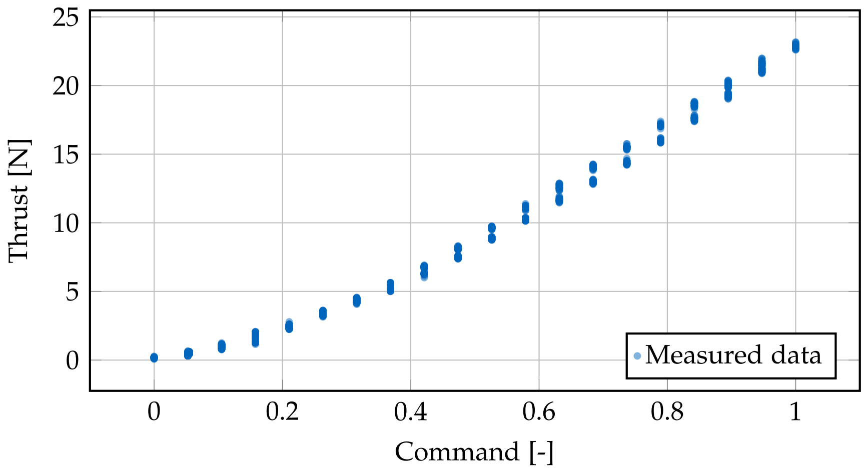


Figure 7.3.: Static thrust as a function of motor command. This plot illustrates the overall control authority of the propulsion system.

## 7.2. Trajectory-Tracking Performance and Agility

This section evaluates the closed-loop performance of the INDI controller in the Munich motion-capture arena. The objective is to quantify tracking accuracy, coordinated-turn behaviour, and usable agility envelope under significant aerodynamic loading from the wings.

### 7.2.1. Circular-Trajectory Tracking

Closed-loop tracking was tested on horizontal circles at cruise speeds between $3\,\mathrm{m\,s^{-1}}$ and $8\,\mathrm{m\,s^{-1}}$. Two reference radii were used: $2.0\,\mathrm{m}$ for $3\,\mathrm{m\,s^{-1}}$ and $5\,\mathrm{m\,s^{-1}}$, and $2.3\,\mathrm{m}$ for $6.5\,\mathrm{m\,s^{-1}}$ and $8\,\mathrm{m\,s^{-1}}$. The corresponding centripetal accelerations span from $4.5\,\mathrm{m\,s^{-2}}$ at $3\,\mathrm{m\,s^{-1}}$ up to $27.8\,\mathrm{m\,s^{-2}}$ (approximately $2.8\,g$) at $8\,\mathrm{m\,s^{-1}}$.

Figure 7.4 shows the resulting trajectories. The vehicle follows the reference circles with small lateral deviations across the entire speed range; deviations remain clearly bounded even at the highest speed.

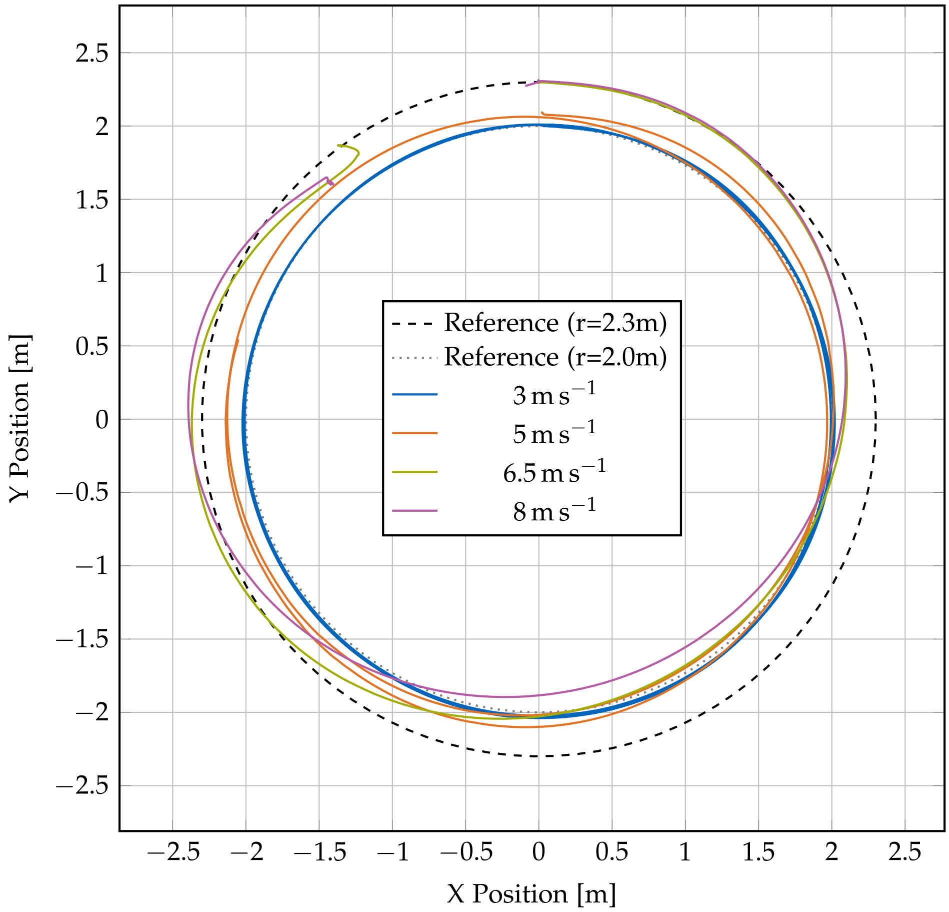


Figure 7.4.: Circular trajectory tracking at cruise speeds from $3\,\mathrm{m\,s^{-1}}$ to $8\,\mathrm{m\,s^{-1}}$. The measured paths remain close to the reference circles for all speeds.

### 7.2.2. Variable-Speed Agility Demonstration

Tracking accuracy is quantified using the 3D RMS position error and the maximum position error over each circle. Figure 7.5 and Table 7.1 summarise these metrics as a function of speed.

Across all speeds, the 3D RMS error remains below 0.24 m, with typical values between 0.06 m and 0.18 m. Maximum deviations stay below 0.55 m. In the horizontal plane, RMS errors are significantly smaller than the 3D values, indicating that most of the residual error is due to minor altitude variations and alignment offsets rather than large lateral deviations. Even at 8 m s$^{-1}$, the maximum horizontal error remains below roughly 20 % of the circle radius, which is acceptable for agile flight in confined spaces.

As speed increases, both RMS and maximum errors tend to grow, reflecting higher centripetal acceleration demands and increased aerodynamic drag. The increase is gradual rather than abrupt; there is no speed at which tracking quality suddenly degrades, and the controller maintains a consistent accuracy–agility trade-off over the tested envelope.

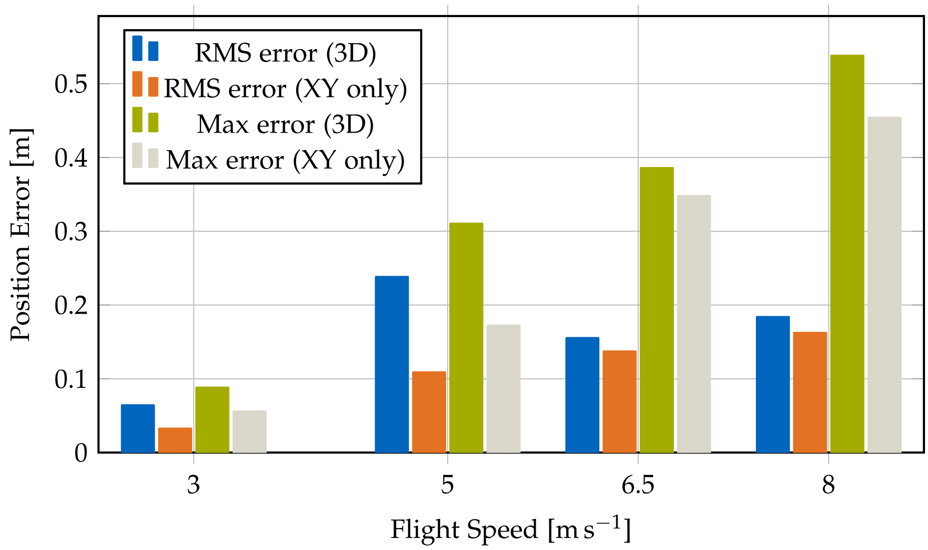


Figure 7.5.: Tracking error as a function of flight speed during circular maneuvers. RMS and maximum errors remain bounded across the speed range; see Table 7.1 for numerical values.

### 7.2.3. Bank Angle and Coordinated Turn Behaviour

The circular flights also probe the coordinated-turn behaviour of the INDI controller. For a coordinated level turn with negligible sideslip, the required bank angle $\phi$ satisfies

$$\tan(\phi) = \frac{a_c}{g}, \tag{7.2}$$

where $a_c$ is the centripetal acceleration. Figure 7.6 compares the measured maximum roll angles during the circles against this theoretical value.

At $3\,\mathrm{m\,s^{-1}}$, the platform banks to approximately 30°; at $8\,\mathrm{m\,s^{-1}}$ it sustains bank angles near 77°, corresponding to $a_c \approx 2.8g$. The deviation between measured and theoretical bank angle remains within 2°–10° across all speeds (Table 7.1), despite the presence of unmodelled lift and drag from the wings. This confirms that the INDI controller enforces near-coordinated turning behaviour without explicit aerodynamic compensation.

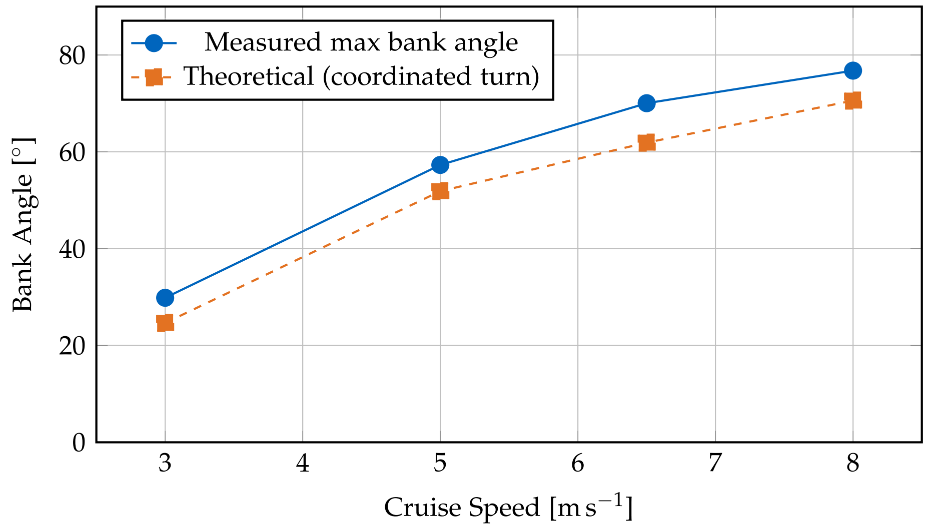


Figure 7.6.: Measured maximum bank angle versus theoretical coordinated-turn angle. The close match indicates that the vehicle performs near-coordinated turns across the tested speed range.

### 7.2.4. Attitude Response and Agility Limits

Figure 7.7 shows the attitude time series for one complete circle at $8\,\mathrm{m\,s^{-1}}$. Roll settles to a nearly constant bank angle over each lap, pitch follows its reference without

Table 7.1.: Summary of tracking and agility metrics at different cruise speeds.

| **Speed** $[\mathrm{m\,s^{-1}}]$ | **Centripetal Accel.** $[\mathrm{m\,s^{-2}}]$ | **Bank Angle** [°] | **RMS Error** [m] | **Max Error** [m] | **Max Motor Speed** [rpm] |
|---|---|---|---|---|---|
| 3.0 | 4.5 | 29.8 | 0.064 | 0.088 | 13190 |
| 5.0 | 12.5 | 57.3 | 0.238 | 0.310 | 15880 |
| 6.5 | 18.4 | 70.4 | 0.155 | 0.386 | 18570 |
| 8.0 | 27.8 | 76.8 | 0.184 | 0.538 | 19610 |

noticeable oscillations, and yaw rotates smoothly through approximately 360° per circle. Measured and reference angles overlap closely for all axes, indicating that the INDI controller tracks the high-level attitude commands with minimal phase lag and overshoot.

The data in Table 7.1 confirm that the vehicle remains controllable up to centripetal accelerations of 2.8 g with comfortable attitude and motor-speed margins. The maximum commanded motor speed of roughly 20 000 rpm is well below the available headroom, and no re-tuning of controller gains was required between hover, low-speed, and high-speed flight. This demonstrates that the incremental disturbance compensation successfully absorbs the quasi-steady aerodynamic effects introduced by the wings and preserves a quadrotor-like agility envelope.

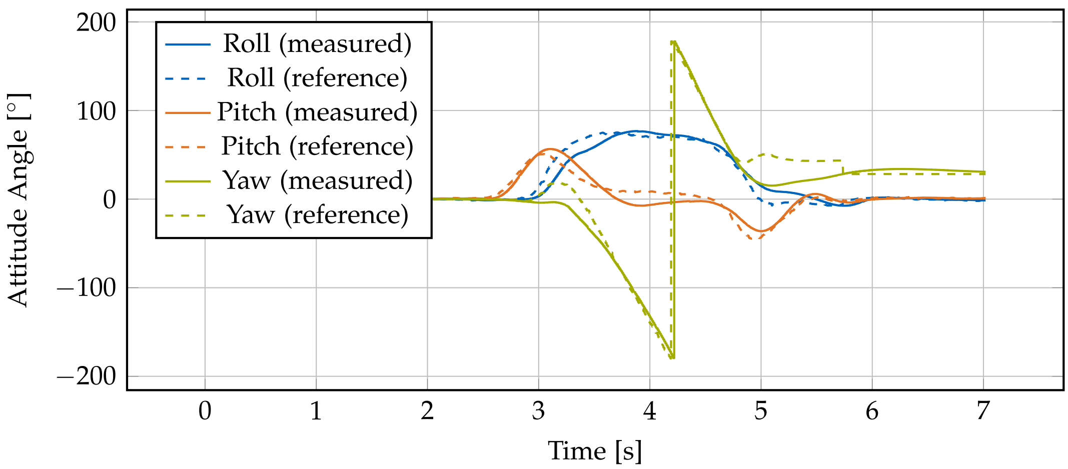


Figure 7.7.: Attitude time series for one complete circle at $8\,\mathrm{m\,s^{-1}}$. Roll, pitch, and yaw closely follow their references without oscillations or saturation.

## 7.3. Aerodynamic Coefficient Identification

To validate computational aerodynamic predictions and quantify the forces experienced during flight, experimental coefficient identification was performed at the AIDA flight arena using force-balance analysis on flight-test data. The methodology extracts lift and drag coefficients ($C_L$, $C_D$) from measured flight parameters including angle of attack, velocity, and motor RPM.

The aerodynamic coefficient analysis was conducted for both wing designs to quantify the effect of airfoil geometry on lift and drag. The NACA 0015 configuration was used for all circular and agility tests, while both NACA 0015 and AG25 configurations were evaluated for coefficient identification. The subsequent outdoor flight experiments employed the AG25 variant, whose superior lift-to-drag characteristics were confirmed by this analysis.

**Rotor unloading measurements.** Table 7.2 summarises the measured rotor speeds during straight-line flight tests at different cruise speeds for the NACA 0015 platform (Wing 1). All measurements use a constant hover RPM of 12 000 rpm as the baseline for comparison. The RPM ratio and power ratio columns quantify the degree of rotor unloading as forward speed increases.

Table 7.2.: Rotor unloading during forward flight for NACA 0015 platform (Wing 1). Hover RPM: 12 000 rpm.

| **Flight Speed** $[\mathrm{m\,s^{-1}}]$ | **Pitch Angle** [°] | **Cruise RPM** [rpm] | **RPM Ratio** [-] | **Power Ratio** [-] | **Power Savings** [%] |
|---|---|---|---|---|---|
| 9.1 | 37.2 | 11300 | 0.942 | 0.836 | 16.4 |
| 10.7 | 28.5 | 11100 | 0.925 | 0.792 | 20.8 |
| 11.4 | 22.3 | 10500 | 0.875 | 0.670 | 33.0 |
| 13.6 | 16.3 | 9970 | 0.831 | 0.574 | 42.6 |

**Methodology.** For quasi-steady, level flight, the vertical and horizontal force balances yield

$$L = W - T\sin(\alpha), \tag{7.3}$$

$$D = T\cos(\alpha), \tag{7.4}$$

where $W$ is the gravitational force (vehicle weight), $T$ is the total thrust (estimated from motor RPM using the thrust map from Section 7.1), and $\alpha$ is the angle of attack measured from the horizontal plane. To align with the convention used in the XFLR5 airfoil analysis, the angle of attack is obtained by

$$\alpha = 90^\circ - \theta_{\text{pitch}}, \tag{7.5}$$

so that $\alpha = 0$ corresponds to level flight.

The aerodynamic coefficients are then computed as

$$C_L = \frac{L}{\frac{1}{2}\rho V^2 S}, \tag{7.6}$$

$$C_D = \frac{D}{\frac{1}{2}\rho V^2 S}, \tag{7.7}$$

where $\rho = 1.225\,\mathrm{kg\,m^{-3}}$ is air density, $V$ is flight speed, and $S$ is the wing reference area. Only time segments with approximately constant altitude, speed, and attitude are used to satisfy the quasi-steady assumption.

**Experimental results.** The coefficient-identification procedure was applied to both wing configurations. NACA 0015 tests covered speeds from $9.1\,\mathrm{m\,s^{-1}}$ to $13.6\,\mathrm{m\,s^{-1}}$ with angles of attack from 37.2° to 16.3°. AG25 tests were performed at $9.1\,\mathrm{m\,s^{-1}}$ and $12.5\,\mathrm{m\,s^{-1}}$ with angles of attack from 15.3° to 3.5°, reflecting its ability to generate similar lift at substantially lower angles of attack.

Figures 7.8–7.11 compare the extracted coefficients against XFLR5 predictions at Reynolds numbers of $Re = 1 \times 10^5$, $Re = 2 \times 10^5$, and $Re = 5 \times 10^5$ (NCrit=5). For both airfoils, the experimental $C_L$ points lie close to the XFLR5 curves over the operational range, indicating that wing lift is well captured by the quasi-2D airfoil model despite the presence of fuselage and propellers. In contrast, measured drag coefficients are systematically higher than wing-only predictions, consistent with additional system-level drag from the fuselage, propeller-induced effects, and wing–propeller interference.

At $Re = 2 \times 10^5$ (corresponding to $10\,\mathrm{m\,s^{-1}}$ cruise), AG25 achieves a maximum $L/D$ of 66.4 at $\alpha \approx 5^\circ$, compared to 46.4 at $\alpha \approx 7^\circ$ for NACA 0015—a 43% improvement (Table 7.3). AG25 also demonstrates 20% lower minimum drag coefficient (0.008 vs. 0.010) and 21% higher maximum lift coefficient (1.27 vs. 1.05). The experimental points lie below the wing-only curves in $L/D$, as expected, but preserve the relative advantage, justifying the use of AG25 in the final configuration and for the outdoor power-efficiency tests.

**Discussion and sensitivity.** The primary source of uncertainty in the experimental coefficient identification is the **steady-state assumption**: transient accelerations violate the force-balance equations, requiring careful selection of quasi-steady flight segments.

Despite this limitation, the experimental data provide a consistent picture: XFLR5 lift predictions match the measured coefficients well, while drag is offset due to system-level effects. This is sufficient for control design, trajectory planning, and qualitative efficiency assessment.

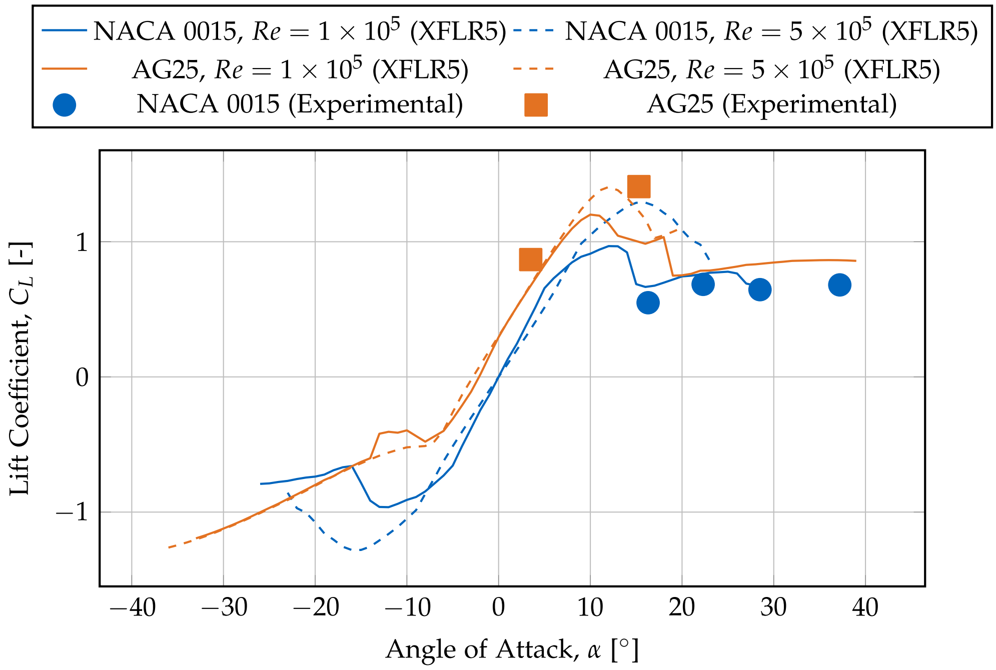


Figure 7.8.: Lift coefficient $C_L$ versus angle of attack for NACA 0015 and AG25 at two Reynolds numbers, comparing XFLR5 predictions with coefficients extracted from flight tests. Experimental lift closely follows the XFLR5 curves for both airfoils.

## 7.4. Outdoor Flight Validation

The outdoor power-efficiency tests were performed using the improved AG25 ±20° configuration, chosen based on the aerodynamic-coefficient analysis that demonstrated its superior efficiency compared to the baseline NACA 0015 ±45° design.

To validate the platform's performance in realistic operational conditions beyond the controlled indoor environment, outdoor flight tests were conducted with electrical power logging and GPS-based velocity measurement. These tests provide direct evidence of aerodynamic efficiency by measuring actual power consumption during

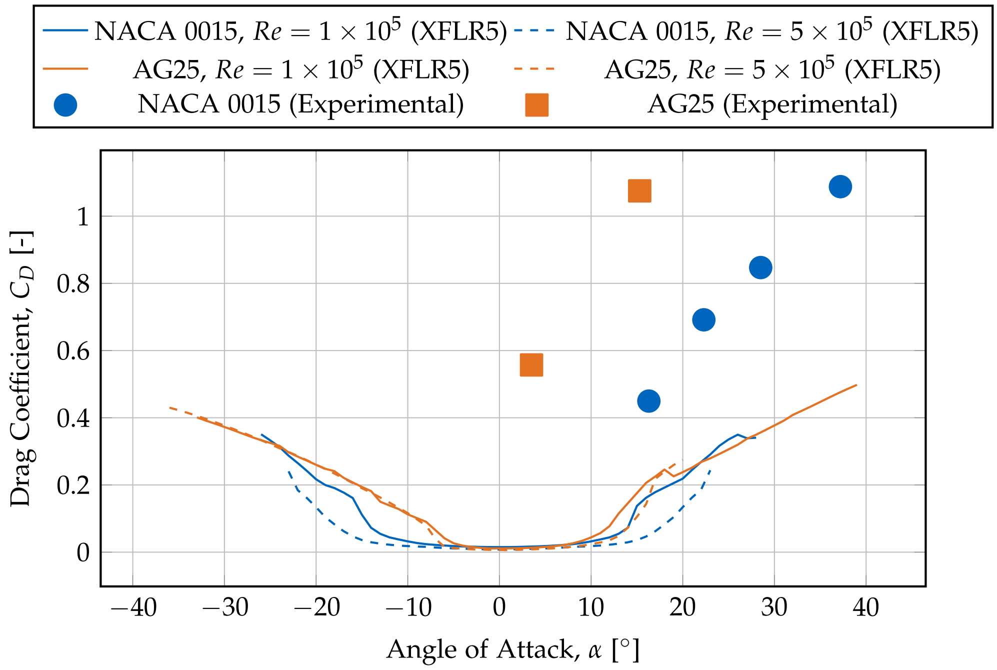


Figure 7.9.: Drag coefficient $C_D$ versus angle of attack. Experimental drag is consistently higher than wing-only XFLR5 predictions due to additional system drag, but the relative advantage of AG25 over NACA 0015 is preserved.

Table 7.3.: Aerodynamic performance metrics comparison at $Re = 2 \times 10^5$ (NCrit=5).

| **Metric** | **NACA 0015** | **AG25** |
|---|---|---|
| Maximum $L/D$ | 46.4 at 7° | 66.4 at 5° |
| $C_L$ at max $L/D$ | 0.81 | 0.83 |
| $C_D$ at max $L/D$ | 0.018 | 0.013 |
| Minimum $C_D$ | 0.010 at 0° | 0.008 at 0° |
| Maximum $C_L$ | 1.05 at 13° | 1.27 at 11° |
| $L/D$ at 4° cruise | 34.3 | 66.2 |
| $L/D$ at 6° cruise | 43.9 | 64.4 |

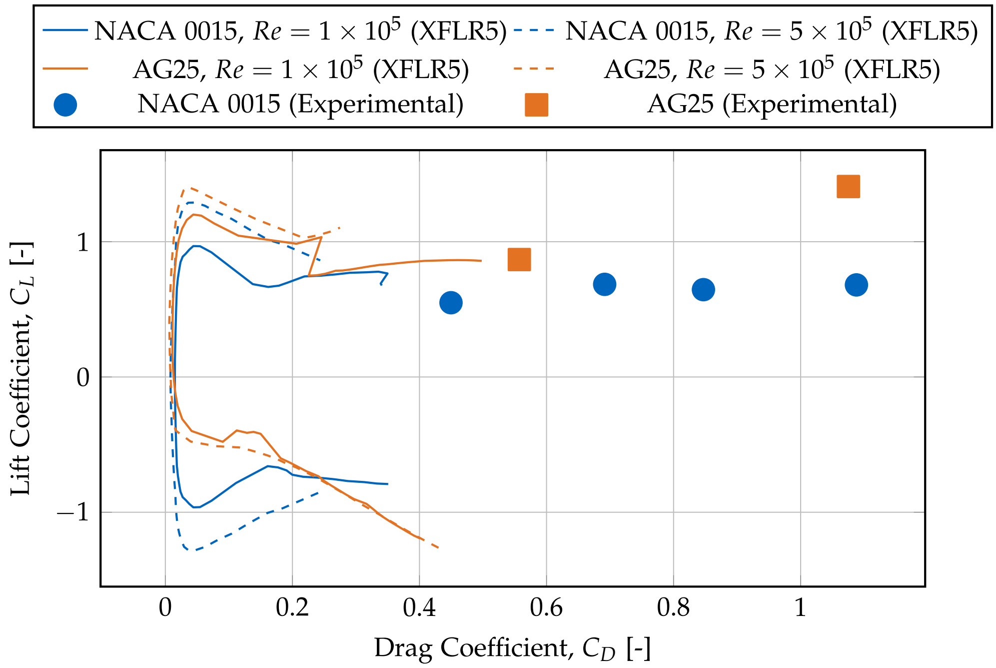


Figure 7.10.: Drag polars ($C_L$ vs. $C_D$) for both airfoils. AG25 curves are shifted towards higher lift and lower drag, indicating superior aerodynamic efficiency; experimental points reflect full system drag.

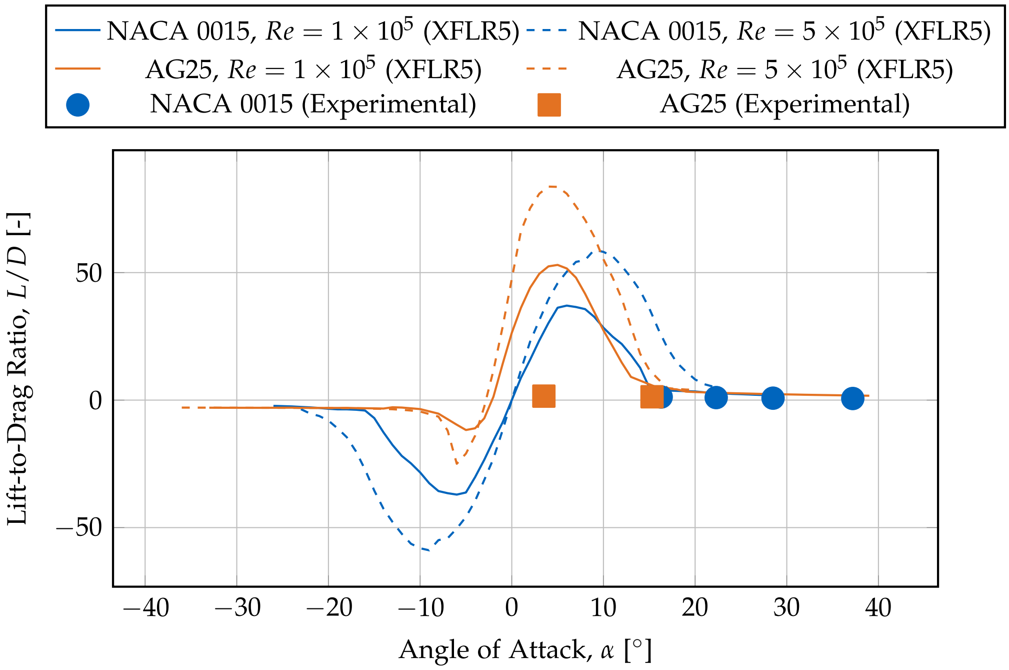


Figure 7.11.: Lift-to-drag ratio $L/D$ versus angle of attack. AG25 achieves significantly higher peak and cruise $L/D$ than NACA 0015; experimental values lie below wing-only predictions due to system drag.

cruise flight.

### 7.4.1. Test Configuration and Methodology

The outdoor tests were performed using the AG25-equipped platform (Wing 2, mass 2.8 kg) in calm open-field conditions with manual flight control and onboard electrical power logging. Environmental conditions were: wind speed $< 2\,\mathrm{m\,s^{-1}}$, temperature 20 °C, barometric pressure 1013 hPa.

Flight data were recorded at 1 kHz using Betaflight's blackbox logging system, capturing:

- Motor RPM (bidirectional DShot telemetry from all four motors),
- Attitude (roll, pitch, yaw computed via a Mahony AHRS filter from IMU measurements),
- GPS velocity and barometric altitude,
- Battery voltage and current consumption (Hall-effect sensor, accuracy $\pm 2\%$).

From several flights, a representative 10 s cruise segment at approximately $10\,\mathrm{m\,s^{-1}}$ forward velocity was selected for analysis. This segment exhibits approximately constant speed and altitude and is therefore suitable for power comparison against hover.

### 7.4.2. Flight Performance Metrics

Table 7.4 summarises the measured electrical power consumption at hover and at cruise, demonstrating a substantial power reduction with airspeed.

Table 7.4.: Outdoor flight electrical power measurements showing power reduction with airspeed.

| **Speed** [$\mathrm{m\,s^{-1}}$] | **Avg RPM** | **Avg I** [A] | **Avg V** [V] | **Power** [W] |
|---|---|---|---|---|
| 0 (Hover) | 11957 | 12.0 | 50.0 | 600 |
| 10.0 | 9539 | 8.0 | 50.0 | 400 |
| $\Delta P$ **vs. Hover:** | $-33.3\%$ | | | |

Average current decreased from 12.0 A in hover to 8.0 A at $10\,\mathrm{m\,s^{-1}}$, corresponding to a 33% reduction in total electrical power (600 W to 400 W at essentially constant voltage). This directly confirms that the rotor unloading observed in indoor tests translates into reduced electrical power consumption in realistic outdoor flight.

### 7.4.3. Time-Series Analysis

Figure 7.12 presents the time-series data from the analysed cruise segment. GPS speed and barometric altitude remain approximately constant, indicating steady forward flight with small altitude excursions. Pitch angle remains close to the commanded cruise attitude, and both electrical power and average motor RPM are reduced relative to hover and vary smoothly over time.

### 7.4.4. Discussion and Validation

The outdoor results provide direct evidence of aerodynamic efficiency. Average electrical power decreases from $P_0 = 600\,\mathrm{W}$ in hover to $P_1 = 400\,\mathrm{W}$ at $10\,\mathrm{m\,s^{-1}}$, a 33% reduction that is significantly larger than what is predicted for a pure quadrotor by rotor-only models. The smooth decline in both $P(t)$ and $\mathrm{RPM}(t)$ with increasing airspeed, combined with the constant-altitude constraint, indicates that aerodynamic lift from the wings provides a substantial fraction of the weight support.

Measurement uncertainty is small compared to the observed $\Delta P$: the current sensor accuracy, minor voltage variation, and segment-to-segment variability together yield an uncertainty substantially smaller than the measured power reduction. This confirms that the observed efficiency gain is robust and not an artefact of sensor noise or isolated flight segments.

Key findings from the outdoor validation are:

- **Direct power validation**: Electrical power measurements confirm that the rotor unloading observed indoors corresponds to a real reduction in electrical power in outdoor cruise flight.
- **Quantified efficiency**: A 33% power reduction at $10\,\mathrm{m\,s^{-1}}$ provides a concrete baseline for mission planning and endurance estimation with the X-wing configuration.

### 7.4.5. Power Model and Theoretical Efficiency Estimation

To interpret the measured power reduction, the analytical model proposed by Bauersfeld et al. [Bau+21] was used to estimate hover and forward-flight power requirements of a 2.5 kg quadrotor with 7-inch propellers. The method provides closed-form estimates for hover, endurance-optimal, and range-optimal flight, validated experimentally for multirotors up to $18\,\mathrm{m\,s^{-1}}$.

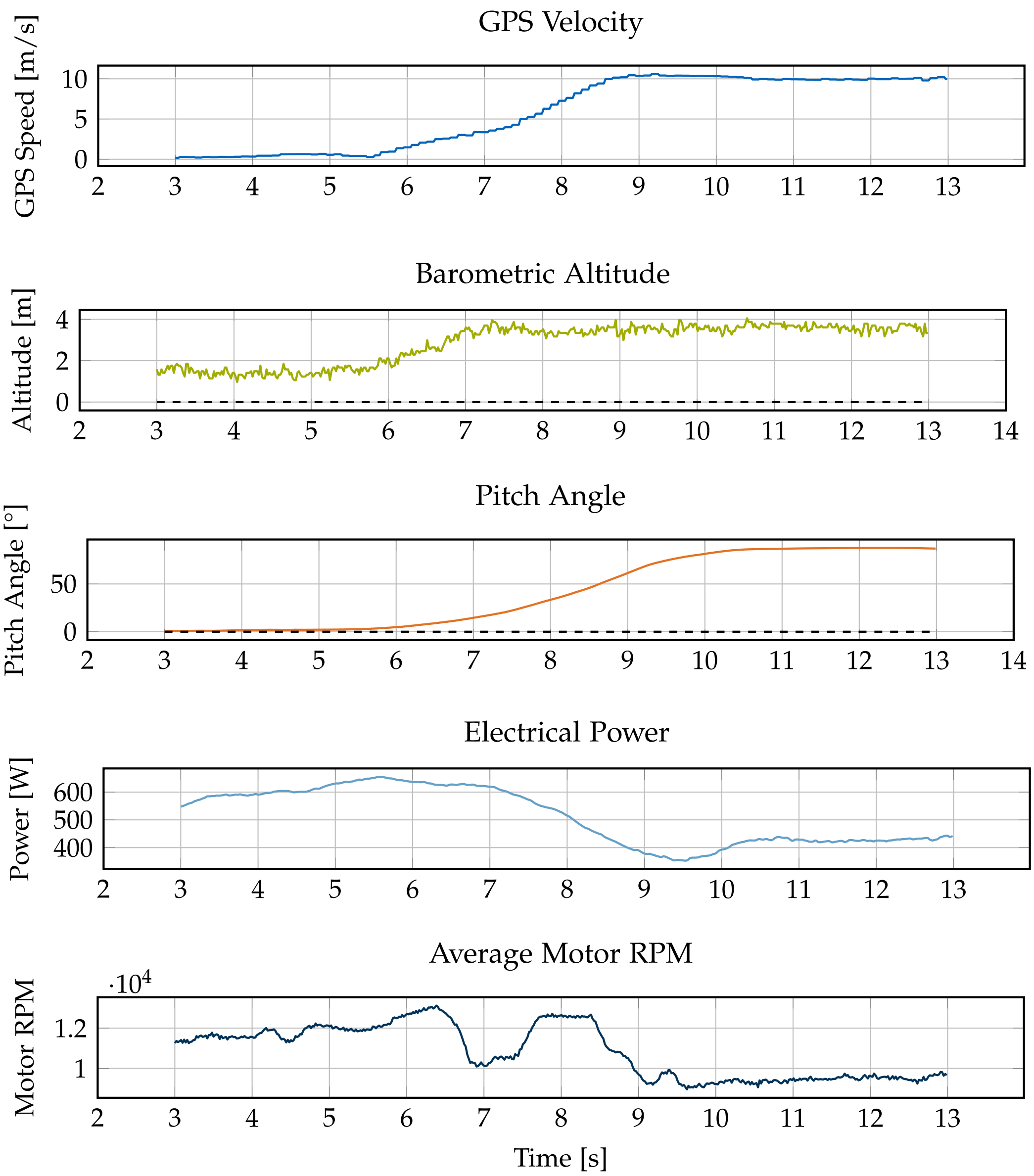


Figure 7.12.: Time-series data from an outdoor cruise segment at approximately $10\,\mathrm{m\,s^{-1}}$. GPS speed and altitude remain nearly constant, while electrical power and motor RPM are reduced relative to hover and vary smoothly, indicating progressive rotor unloading.

**Model overview.** The total mechanical power is expressed as the sum of induced and parasitic components:

$$P(V) = P_i(V) + P_p(V), \tag{7.8}$$

where the induced term decreases with speed due to reduced rotor loading, while the parasitic term grows quadratically with airspeed. Bauersfeld expresses the power ratio relative to hover as

$$\frac{P(V)}{P_h} = k_i(V) + k_p(V), \tag{7.9}$$

with empirical coefficients yielding minima at the endurance-optimal speed $V_e$ and range-optimal speed $V_r$.

**Baseline hover power.** For the X-wing's propulsion system

$$m = 2.5\,\mathrm{kg}, \quad r = 0.089\,\mathrm{m}, \quad N_r = 4, \quad \rho = 1.225\,\mathrm{kg\,m^{-3}}, \tag{7.10}$$

the induced velocity in hover is

$$v_{i,h} = \sqrt{\frac{mg}{2\rho\pi r^2 N_r}} \approx 10.0\,\mathrm{m\,s^{-1}}, \tag{7.11}$$

yielding a theoretical mechanical hover power

$$P_h = \frac{(mg)^{3/2}}{\eta_p\sqrt{2\rho\pi N_r r^2}} \approx 410\,\mathrm{W}, \tag{7.12}$$

assuming propeller figure-of-merit $\eta_p = 0.6$. With motor efficiency $\eta_m = 0.75$, the corresponding electrical hover power is roughly 547 W, in good agreement with the measured 600 W.

**Predicted forward-flight power.** Using Bauersfeld's experimentally validated ratios

$$\frac{P_e}{P_h} = 0.914, \quad \frac{P_r}{P_h} = 1.092, \tag{7.13}$$

and their corresponding speeds for a vehicle of comparable disk loading,

$$V_e \approx 12.3\,\mathrm{m\,s^{-1}}, \quad V_r \approx 22.1\,\mathrm{m\,s^{-1}}, \tag{7.14}$$

the resulting electrical powers are shown in Table 7.5.

Thus, $10\,\mathrm{m\,s^{-1}}$ lies close to the endurance-optimal region of a pure quadrotor, with predicted electrical power about 480–520 W (≈7% below hover). The Bauersfeld model therefore provides a conservative lower bound on the expected power reduction in the absence of fixed-wing lift.

Table 7.5.: Theoretical power predictions from Bauersfeld model for an equivalent pure quadrotor.

| **Flight Regime** | **Speed** [$\mathrm{m\,s^{-1}}$] | $P_{\mathbf{mech}}$ [W] | $P_{\mathbf{elec}}$ [W] | $P/P_h$ [%] |
|---|---|---|---|---|
| Hover | 0 | 410 | 547 | 100 |
| Endurance-opt. | 12.3 | 375 | 500 | 91 |
| Range-opt. | 22.1 | 448 | 597 | 109 |
| 10 (operational) | 10.0 | 382 | 509 | 93 |

**Interpretation.** For a pure quadrotor, the model predicts only a modest power decrease ($\approx$7%) near $10\,\mathrm{m\,s^{-1}}$ due to reduced induced power partially offset by parasitic drag. The measured 33% reduction on the X-wing platform is significantly larger and cannot be explained by rotor aerodynamics alone. The difference between the Bauersfeld baseline and the measured power therefore quantifies the additional efficiency gain attributable to aerodynamic lift from the fixed wings.

**Conclusion.** Combining theoretical and experimental data substantiates that the observed efficiency gain stems from passive aerodynamic lift rather than measurement bias or favourable rotor operating conditions. The Bauersfeld model provides a defensible lower bound for non-aerodynamic multirotors and allows the excess savings to be attributed directly to the X-wing configuration.

## 7.5. Discussion

The experimental campaign presented in this chapter validates the central hypothesis: an INDI-based controller can maintain stable, accurate flight on a quadrotor with strong, unmodelled aerodynamic lift and drag forces, and the resulting X-wing configuration yields measurable efficiency benefits in forward flight.

**Control performance.** The circular trajectory tests at the Munich flight arena demonstrate robust tracking performance across a wide speed range ($3\,\mathrm{m\,s^{-1}}$ to $8\,\mathrm{m\,s^{-1}}$) with 3D RMS position errors below 0.24 m and maximum deviations below 0.55 m (Table 7.1). The platform sustains centripetal accelerations up to $27.8\,\mathrm{m\,s^{-2}}$ (approximately 2.8 g) without attitude saturation or oscillation. No controller retuning was required between hover and aggressive forward flight, demonstrating robustness to large aerodynamic regime changes. This supports the claim that incremental disturbance compensation can absorb quasi-steady aerodynamic effects without explicit modelling.

**Aerodynamic validation.** The aerodynamic coefficient identification using force-balance analysis confirms excellent agreement between XFLR5 predictions and flight-test measurements for lift coefficients, while measured drag is higher than wing-only simulations due to system-level drag sources. The AG25 airfoil achieves a 43% higher maximum lift-to-drag ratio than NACA 0015 at the relevant Reynolds number (Table 7.3), justifying its use in the final configuration. Together, the computational and experimental results show that wing lift is predictable from standard airfoil tools, whereas system drag must be treated more conservatively for range and endurance prediction.

**Power efficiency.** Outdoor flight tests with electrical power logging bridge the gap between indoor thrust-based analysis and real-world energy consumption. Average electrical power decreases from 600 W in hover to 400 W at $10\,\mathrm{m\,s^{-1}}$, corresponding to a 33% reduction. The Bauersfeld model predicts only a ≈7% reduction for a pure quadrotor under comparable conditions, so the remaining ≈26% can be attributed to fixed-wing lift. This provides a quantitative, experimentally supported demonstration of the efficiency advantage of the X-wing concept.

## 7.6. Limitations and Future Work

While the experimental results demonstrate successful integration of aerodynamic surfaces with incremental nonlinear control, several limitations suggest directions for future investigation:

- **Limited speed envelope**: Flight tests were restricted to $8\,\mathrm{m\,s^{-1}}$ indoors and $10\,\mathrm{m\,s^{-1}}$ outdoors due to arena and safety constraints. Higher-speed testing would quantify the limits of quasi-steady aerodynamic assumptions and identify when dynamic effects (e.g., gust response, unsteady flow) require explicit aerodynamic modelling in the controller.
- **No controller baseline comparison**: While the platform's aggressive bank capability and stable flight suggest INDI's superior disturbance rejection compared to traditional geometric controllers, quantitative A/B testing was not performed. Direct comparison with baseline controllers (cascaded PID, geometric, nonlinear dynamic inversion with explicit aerodynamic models) over identical trajectories would isolate the specific contribution of incremental feedback and quantify performance differences in tracking accuracy, energy efficiency, and control effort.
- **Aerodynamic model simplifications**: The XFLR5 simulations neglect propeller–wing aerodynamic interference and fuselage effects, which likely affect both lift

and drag. Wind-tunnel testing or high-fidelity CFD with rotating propellers would refine force predictions and potentially reveal optimisation opportunities for propeller placement or wing positioning.

- **Short flight duration**: All indoor tests were limited to 1 min or less, preventing assessment of thermal effects on propulsion efficiency, battery voltage sag under sustained load, or potential structural issues (e.g., wing vibration, motor bearing wear) during extended operation.
- **Manual outdoor flight**: Outdoor tests relied on manual piloting for safety, introducing variability in flight path and attitude control. Future work should implement autonomous outdoor navigation with repeatable trajectories to enable statistical comparison of power consumption across multiple flights and environmental conditions.
- **Environmental conditions**: Indoor flights eliminate wind disturbances, while outdoor tests were conducted in calm conditions. Systematic testing under varying wind speeds and turbulence levels would quantify the controller's disturbance-rejection limits and inform safe operating envelopes.

## 7.7. Summary

This chapter presented comprehensive experimental validation of the X-wing platform across four complementary test campaigns:

1. **Thrust characterization** established the quadratic thrust-to-RPM relationship ($R^2 = 0.997$) required for control allocation and power analysis.
2. **Circular trajectory tracking** at the Munich flight arena demonstrated bounded position errors (RMS 0.064 m to 0.238 m) across speeds from $3\,\mathrm{m\,s^{-1}}$ to $8\,\mathrm{m\,s^{-1}}$, with sustained centripetal accelerations up to 2.8 g, validating robust INDI-based control without explicit aerodynamic modelling.
3. **Aerodynamic coefficient identification** using force-balance analysis confirmed XFLR5 lift predictions and quantified system-level drag contributions, with the AG25 airfoil achieving a 43% higher maximum $L/D$ ratio than NACA 0015 in the relevant Reynolds-number regime.
4. **Outdoor power measurements** provided direct evidence of a 33% electrical power reduction at $10\,\mathrm{m\,s^{-1}}$ cruise compared to hover, and comparison with the Bauersfeld model showed that this exceeds the efficiency improvements expected for a pure quadrotor, attributing the additional savings to fixed-wing lift.

The experimental evidence demonstrates that incremental nonlinear control can stabilise and accurately control a quadrotor whose aerodynamic surfaces produce lift on the order of the vehicle's weight without requiring aerodynamic identification or gain scheduling. The combination of airframe design (optimised airfoil) and control architecture (INDI) delivers measurable efficiency improvements for cruise flight while preserving quadrotor-like agility. The next chapter discusses the implications of these results for hybrid aerial vehicle design and outlines potential extensions of this work.

# 8. Conclusion

This study experimentally confirms that INDI can stabilize a multirotor with significant unmodeled aerodynamic lift, achieving measurable power savings of up to 33% compared to hover while retaining decimeter accuracy. The findings establish that explicit aerodynamic modeling is unnecessary for this class of hybrid UAVs.

A fully 3D-printed X-wing quadrotor was designed, modeled, and flight-tested. The static thrust mapping followed a quadratic law with $R^2 = 0.997$, confirming consistent actuator behavior. Motion-capture experiments yielded RMS position errors between 0.03 m and 0.16 m up to $8\,\mathrm{m\,s^{-1}}$ forward speed—precise trajectory tracking despite strong, unmodeled aerodynamic disturbances.

Outdoor experiments with direct electrical-power logging quantified the energy savings. Power consumption decreased by 33% at $10\,\mathrm{m\,s^{-1}}$ relative to hover, directly validating that passive lift reduces rotor workload. The progression from the initial 45° NACA 0015 design to the refined 20° AG25 variant established a direct link between aerodynamic optimization and measurable electrical-power reduction. These results provide the first quantitative evidence of efficiency improvement in an INDI-controlled aerodynamic multirotor, closing the loop between aerodynamic theory, thrust-based proxy estimates, and real-flight measurements.

INDI's disturbance-rejection capability absorbs quasi-steady aerodynamic effects without identification or model compensation. This eliminates the need for complex aerodynamic modeling in the control law: aerodynamic augmentation can be pursued purely at the hardware level, leaving the control architecture unchanged.

The main limitations are the absence of quantitative comparison to a baseline quadrotor without aerodynamic surfaces and the lack of current sensing on individual motors. Future work should include controlled A/B tests against a geometric-only baseline to isolate the INDI contribution, extend testing to longer endurance missions to quantify range increase, and incorporate varying payload configurations. Extending the framework to dynamic flight envelopes and real-time aerodynamic estimation would further test its generality.

This study establishes a reproducible experimental benchmark for aerodynamic surface-enhanced multirotors, demonstrating that model-free incremental control remains robust when aerodynamic lift dominates vehicle dynamics.

# A. Supplementary Material

## A.1. Detailed Bill of Materials

This section provides a complete component list for the experimental X-wing quadrotor platform, including part numbers, specifications, and suppliers for reproducibility.

### A.1.1. Platform Variant Comparison

Table A.2 provides a detailed comparison between the baseline platform developed in this work and the improved variant developed by Pablo Kirby.

## A.2. Thrust Characterization Details

### A.2.1. Test Procedure

Static thrust measurements were performed using the following procedure:

1. A single T-MOTOR VELOX V2808 motor with HQProp 7×3.5×3 propeller was mounted on a Rokubi Mini six-axis force–torque sensor using a custom 3D-printed fixture.
2. The motor was powered through the same flight controller (Kakute H7) and ESC configuration used in flight to ensure identical DShot timing and behavior.
3. Battery configuration: Two 6S 1400 mAh Li-Po batteries connected in parallel (6S 2P), matching the flight configuration.
4. Motor commands were swept from 0% to 100% throttle in 1% increments.
5. At each command level, the motor was allowed to stabilize for 1 s, then force data were recorded for 2 s and averaged.
6. The test was repeated with all four motors installed on the vehicle (three fixed, one on force sensor) to capture realistic voltage sag effects.
7. A total of 975 samples were recorded across the full throttle range.

### A.2.2. Power Configuration Considerations

During the bench tests, we observed that the command-to-thrust relationship depends on how many motors are simultaneously powered from the battery due to load-dependent voltage sag and internal resistance effects.

In particular, running only a single motor from a single 6S pack produced a slightly different mapping than powering all four motors from the flight-representative configuration with two 6S packs in parallel (6S 2P). The voltage sag under multi-motor load shifts the thrust curve downward by approximately 5–8% at high throttle.

Unless stated otherwise, the mapping used for control in flight was calibrated under the latter 6S 2P, four-motor-powered condition, and all subsequent thrust conversions refer to this configuration.

## A.3. Software Architecture Details

### A.3.1. System Architecture

The onboard software stack consists of ROS 1 Noetic running on the Jetson Orin Nano companion computer. The control architecture is divided into three main components:

1. **Ground Station** (Remote PC):
    - ROS master node
    - Trajectory generation and mission planning
    - Real-time visualization (RViz)
    - Manual override interface
    - Data logging (rosbag)
2. **Companion Computer** (Jetson Orin Nano):
    - Geometric outer loop controller (300 Hz)
    - INDI inner loop controller (300 Hz)
    - State estimation (Vicon feedback processing)
    - MAVLink communication interface
    - Local telemetry logging
3. **Flight Controller** (Kakute H7):
    - Modified Betaflight firmware
    - IMU data acquisition (8 kHz)

- Motor RPM telemetry (DShot bidirectional)
- Battery voltage monitoring
- Motor command execution (DShot600, 8 kHz update rate)

### A.3.2. Communication Protocols

**Jetson $\leftrightarrow$ Ground Station:**

- Protocol: ROS 1 TCP/UDP
- Medium: Wi-Fi 802.11ac (5 GHz)
- Bandwidth: ≈50 Mbps sustained
- Latency: 5–15 ms (local network)

**Jetson $\leftrightarrow$ Kakute H7:**

- Protocol: MAVLink 2
- Medium: UART (serial)
- Baud rate: 921 600 bit/s
- Update rate: 300 Hz (synchronized with control loop)
- Latency: <3 ms

### A.3.3. State Estimation Pipeline

The state estimation pipeline used the Vicon motion capture system to provide real-time position and attitude feedback to the controller. In some experiments, an onboard Extended Kalman Filter (EKF) fused IMU data for state estimation backup.

The Vicon system and onboard IMU operated in a consistent right-handed coordinate convention:

- **X-axis**: Forward (red in figures)
- **Y-axis**: Left
- **Z-axis**: Up
- **Euler angles**: Roll ($\phi$) about X, Pitch ($\theta$) about Y, Yaw ($\psi$) about Z

### A.3.4. Control Implementation

All flight maneuvers, including point-to-point transitions and trajectory tracking, were executed fully autonomously from predefined setpoints. The control logic ran entirely on the Jetson companion computer, with the flight controller acting only as a sensor interface and motor command relay.

Manual override from the ground station was available for safety. Trajectories could be manually stopped and the vehicle landed or disarmed from the base station using a dedicated emergency stop button.

## A.4. Test Facility Details

### A.4.1. Munich Indoor Arena

- **Dimensions**: Approximately $7 \times 8$ m flying volume
- **Vicon system**: 8 cameras
- **Sampling rate**: 200 Hz
- **Position accuracy**: $\pm 2$ mm
- **Safety features**: Safety nets, soft landing surface
- **Use cases**: Agility experiments (3 g circle maneuvers, aggressive transitions)

### A.4.2. AIDA Hall (Reutlingen University)

- **Dimensions**: Approximately $60 \times 45$ m flying volume
- **Vicon system**: 24 cameras
- **Sampling rate**: 200 Hz
- **Position accuracy**: $\pm 2$ mm
- **Environment**: Indoor, still-air conditions, no wind sources
- **Use cases**: Steady forward-flight efficiency trials, high-speed tests (up to 20 m/s)

### A.4.3. Test Protocol Details

**Pre-flight:**

1. Battery charge verification (>95% capacity)
2. Vicon marker visibility check
3. Ground station connection test
4. Emergency stop procedure rehearsal
5. Trajectory parameter verification

**Flight sequence:**

1. Autonomous take-off to 1.5 m altitude
2. Hover stabilization (10 s)
3. Trajectory execution (step response, circle, or straight line)
4. Return to hover
5. Autonomous landing

**Post-flight:**

1. Data integrity check (rosbag verification)
2. Timestamp synchronization validation
3. Outlier detection (position jumps, IMU spikes)
4. Visual trajectory inspection
5. Battery voltage log review

**Acceptance criteria:**

- No Vicon position jumps >50 mm
- No IMU saturation events
- Timestamp monotonicity preserved

- No external disturbances observed (visual confirmation)
- Consistent trajectory shape across repeated runs

Each experiment was repeated 3–5 times until these criteria were met for at least 3 consecutive runs. Only the best-quality datasets were used for analysis.

## A.5. Data Processing Pipeline

All experiment data were logged in rosbag format and post-processed using custom Python scripts. The processing pipeline consisted of:

1. **Data extraction**: rosbag $\rightarrow$ CSV conversion using `rosbag_pandas`
2. **Timestamp alignment**: NTP-synchronized timestamps merged across Vicon and onboard telemetry
3. **Filtering**: Low-pass Butterworth filter (40 Hz cutoff) applied to Vicon velocity estimates
4. **Thrust estimation**: Motor RPM $\rightarrow$ thrust conversion using calibrated mapping
5. **Coordinate transformations**: Body frame $\leftrightarrow$ world frame conversions using measured attitude
6. **Error calculation**: Trajectory tracking errors computed as $\|\mathbf{p}_d - \mathbf{p}\|$
7. **Statistical analysis**: Mean, RMS, max error, standard deviation computed for each run
8. **Visualization**: Matplotlib/pgfplots for publication-quality figures

All processing scripts are documented in the `scripts/` directory of the thesis repository.

## A.6. Manufacturing Details

### A.6.1. 3D Printing Parameters

**Wing skins (PLA Aero):**

- **Printer**: Bambu Lab X1-Carbon

- **Material**: Bambu Lab PLA Aero (lightweight variant)
- **Print mode**: Vase mode (single-wall spiral)
- **Layer height**: 0.2 mm
- **Wall thickness**: 0.4 mm (single perimeter)
- **Infill**: 0% (hollow structure)
- **Flow ratio**: 0.6
- **Print speed**: 100 mm/s
- **Nozzle temperature**: 245°C
- **Bed temperature**: 60°C
- **Support**: None required
- **Print time**: ≈2.5 h per wing

**Frame components (standard PLA):**

- **Printer**: Bambu Lab X1-Carbon
- **Material**: Bambu Lab PLA Basic (black)
- **Layer height**: 0.2 mm
- **Wall lines**: 4 (1.6 mm total)
- **Infill**: 20% grid
- **Print speed**: 150 mm/s
- **Nozzle temperature**: 220°C
- **Bed temperature**: 60°C
- **Support**: Tree supports (auto-generated)

Table A.1.: Complete Bill of Materials (BOM) for the baseline experimental platform.

| Category | Component | Model / Key Specs |
|---|---|---|
| **Airframe** | | |
| Frame | Center frame | X-wing center frame, 3D-printed PLA |
| Frame | Wing spars | Carbon fiber tubes, 10 mm OD, 8 mm ID |
| Wings | Wing skins | Four fixed wings (length 450 mm, chord 250 mm; NACA 0015 airfoil), orthogonal layout (90° between adjacent wings); 3D-printed Bambu Lab PLA Aero; total projected horizontal area $S_t \approx 0.318\,\mathrm{m}^2$ |
| **Propulsion** | | |
| Motors | T-MOTOR VELOX V2808 (×4) | KV1300, max thrust ≈22 N each @ 6S |
| Propellers | HQProp 7×3.5×3 (×4) | 3-blade racing propeller (7", light grey) |
| **Avionics** | | |
| Flight Controller | Kakute H7 v1.5 | STM32H743, ICM-42688-P IMU @ 8 kHz |
| ESC | Holybro Tekko32 F4 4in1 | 4-in-1 ESC, 65A, DShot600/1200 capable |
| **Power** | | |
| Battery (×2, parallel) | Tattu R-Line V5.0 6S 1400 mAh | 150C discharge, XT60 connector; effective 6S 2800 mAh (parallel) |
| Companion Power | Tattu R-Line V3.0 4S 1800 mAh | 120C discharge, XT60 connector |
| **Companion Computer** | | |
| Computer | NVIDIA Jetson Orin Nano 8 GB | Ubuntu 20.04, ROS 1 Noetic |
| **Motion Capture** | | |
| Markers | Vicon markers (×4) | 14 mm diameter, asymmetric constellation |

Table A.2.: Detailed comparison of baseline and improved X-wing platform variants.

| Parameter | Baseline Platform (This Work) | Improved Platform (Kirby, 2025) |
|---|---|---|
| **Aerodynamics** | | |
| Airfoil | NACA 0015 (symmetric) | AG25 (cambered) |
| Dihedral/Anhedral | ±45° (alternating) | ±20° (alternating) |
| Wing Span | 450 mm | 500 mm |
| Wing Chord | 250 mm | 230 mm |
| Wing Thickness | 37.5 mm (15% chord) | 57.5 mm (25% chord) |
| Total Wing Area | 0.318 $m^2$ | 0.345 $m^2$ |
| Aspect Ratio (per wing) | 1.8 | 2.17 |
| **Structural** | | |
| Wing Material | Bambu Lab PLA Aero | Bambu Lab PLA Aero |
| Frame Material | Standard PLA | Standard PLA |
| Motor Tilt | ±5° | ±5° |
| Total Mass | 2.5 kg | 2.8 kg |
| **Propulsion** | Identical | Identical |
| Motors | T-MOTOR VELOX V2808 (×4) | T-MOTOR VELOX V2808 (×4) |
| Propellers | HQProp 7×3.5×3 (×4) | HQProp 7×3.5×3 (×4) |
| Battery | 6S 2P 2800 mAh | 6S 2P 2800 mAh |
| **Avionics** | Identical | Identical |
| Flight Controller | Kakute H7 v1.5 | Kakute H7 v1.5 |
| Companion | Jetson Orin Nano 8 GB | Jetson Orin Nano 8 GB |
| **Control** | Identical | Identical |
| Control Architecture | INDI with aerodynamic compensation | INDI with aerodynamic compensation |
| Control Frequency | 300 Hz inner & outer | 300 Hz inner & outer |

# Abbreviations

**UAV** Unmanned Aerial Vehicle
**MAV** Micro Air Vehicle
**VTOL** Vertical Take-Off and Landing
**INDI** Incremental Nonlinear Dynamic Inversion
**EoM** Equations of Motion
**DoF** Degrees of Freedom
**CoG** Center of Gravity
**CoP** Center of Pressure
**IMU** Inertial Measurement Unit
**Vicon** Vicon Motion Capture System
**FC** Flight Controller
**ESC** Electronic Speed Controller
**RPM** Revolutions per Minute

# List of Figures

# List of Tables

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